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1 Main Figures

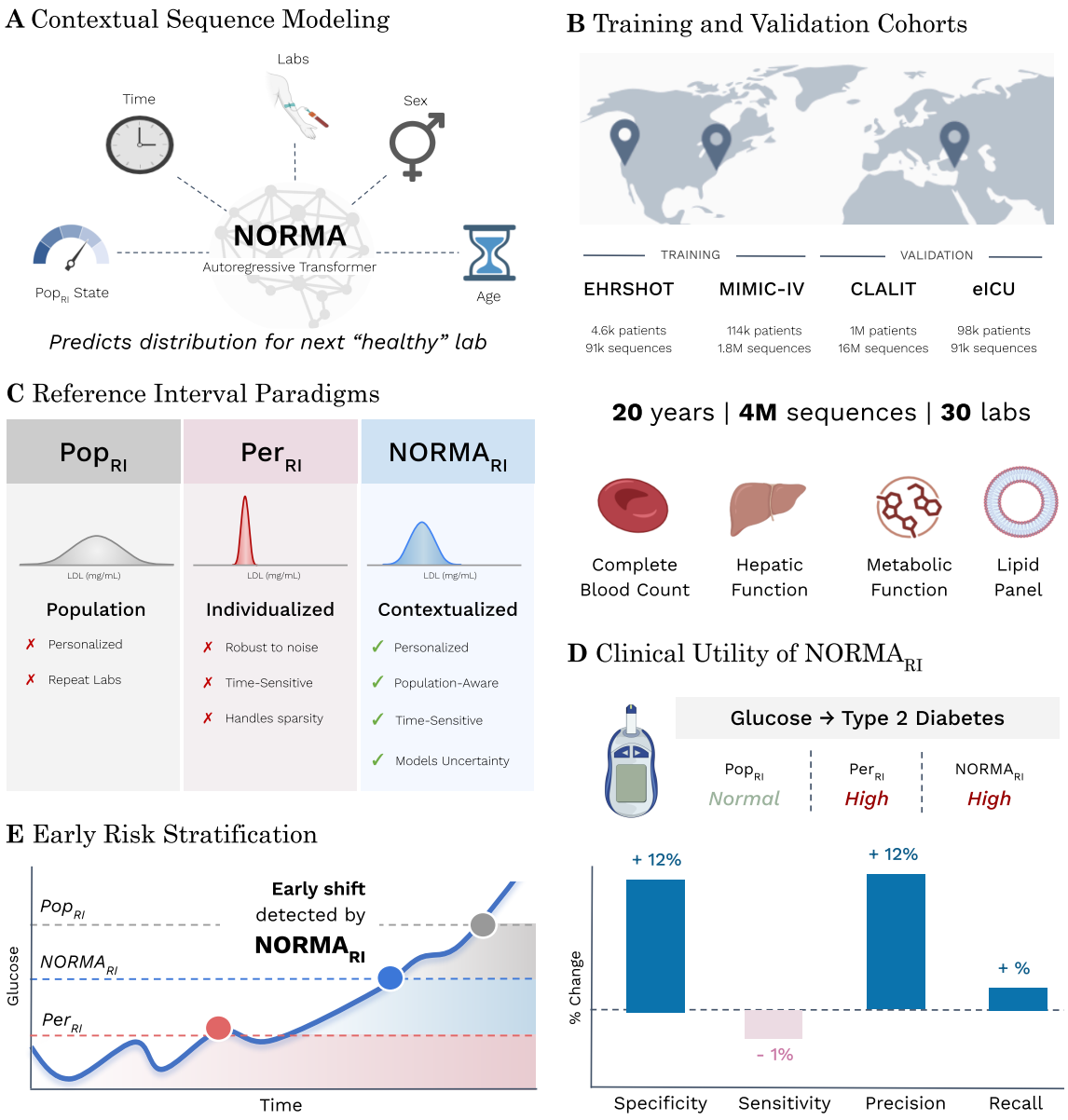

Figure 1: NORMA framework overview. (A) NORMA conditions on laboratory history, time, sex, age, and population reference state to predict the distribution of the next healthy laboratory value. (B) Study design: EHRSHOT and MIMIC-IV (training); Clalit Health Services and eICU Collaborative Research Database (external validation); 20 years, 4 million sequences, 30 analytes across four clinical panels. (C) Comparison of Pop_{RI}, Per_{RI}, and NORMA_{RI}. (D) Illustrative example: a glucose value classified as normal by Pop_{RI} is flagged abnormal by Per_{RI} and NORMA_{RI}, with corresponding change in specificity, sensitivity, and precision for type 2 diabetes prediction. (E) Schematic of early risk stratification by NORMA_{RI} relative to Pop_{RI} and Per_{RI} over time. Created in BioRender.

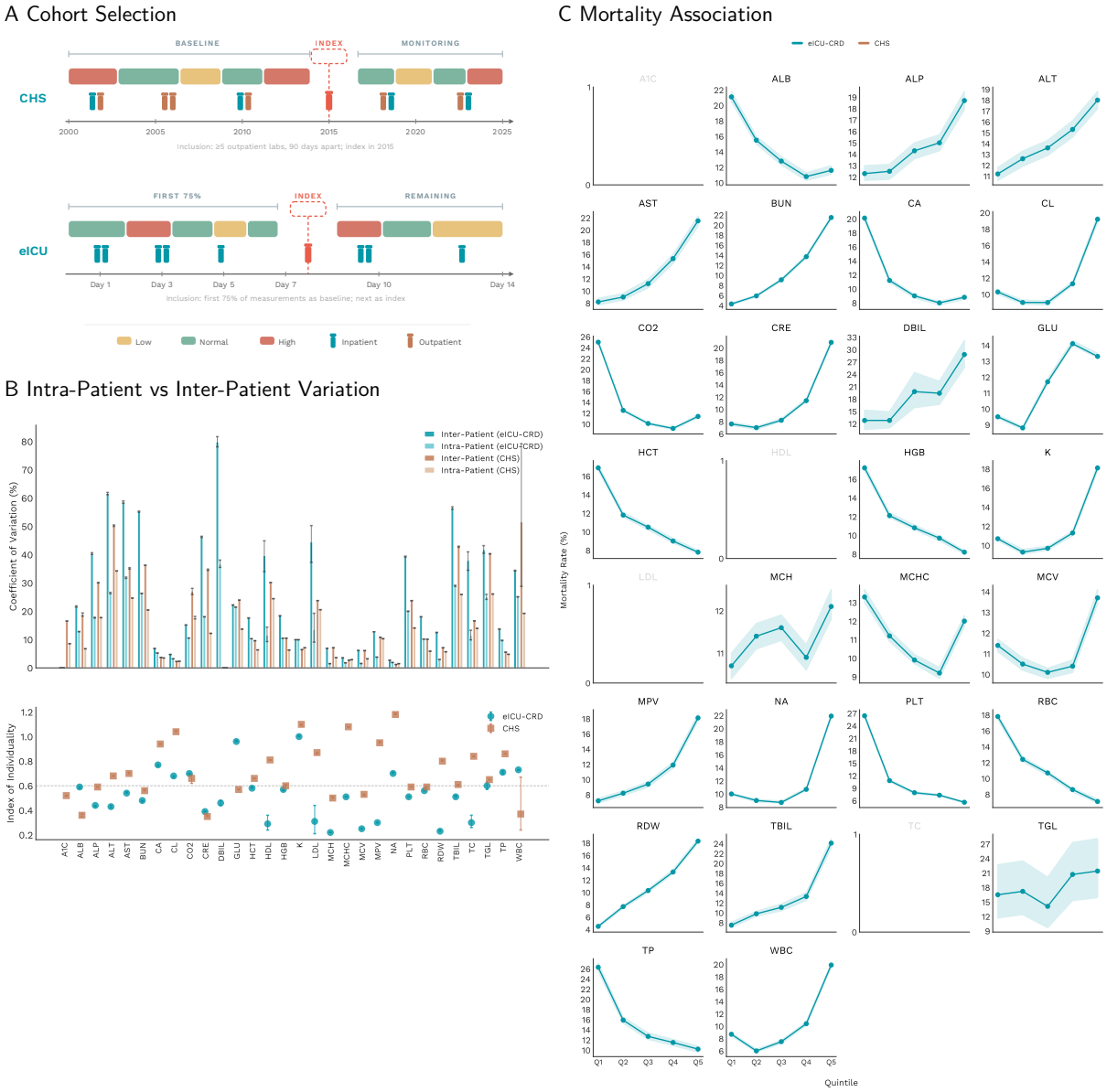

Figure 2: Within-person biological variability and mortality association. A) Cohort Selection. Cohort selection and inclusion criteria for Clalit Health Services and the eICU Collaborative Research Database. **B) Intra-Patient vs Inter-Patient Variation.** Within-person versus between-person coefficient of variation for 30 analytes, with the individuality index shown below. The dashed identity line indicates an individuality index of 0.6. **C) Mortality Association.** Observed mortality rate by quintile of deviation from the NORMA-predicted personalized baseline in Clalit Health Services and the eICU Collaborative Research Database. Shaded bands denote 95% confidence intervals.

Figure 3: NORMA architecture, forecasting accuracy, and sensitivity analysis. A) NORMA Architecture. Model architecture: demographic context, laboratory history, and query tokens are jointly embedded and processed by a causal transformer decoder to produce quantile or Gaussian predictions for the next laboratory value. **B) Forecasting Performance.** Held-out test-set forecasting accuracy (mean absolute error, mean absolute percentage error, and coefficient of determination) for NORMA (quantile) versus ARIMA, population mean, and last-value-carried-forward baselines. Error bars denote 95% bootstrap confidence intervals. **C) Per-Analyte Performance.** Per-analyte coefficient of determination grouped by clinical panel. **D) Sensitivity Analysis.** Sensitivity of prediction interval width (as a percentage of Pop_{RI} width) to history length, forecast horizon, and within-person variability.

Figure 4: Reclassification performance and early risk stratification in the Clalit Health Services cohort. A) Reclassification Rate. Reclassification rate by analyte among Pop_{RI}-normal tests. **B) Hazard Ratios.** Cox hazard ratios for all-cause mortality (adjusted for age and sex); analytes with $p < 0.05$ shown. **C) True Positive Rate vs False Positive Rate. D) Performance Improvement with NORMA.** Change in precision, sensitivity, specificity, and balanced accuracy (NORMA_{RI} versus Per_{RI}); inner label shows the largest gain. **E) Lead Time.** Lead time from NORMA_{RI} flag to first Pop_{RI} flag, by analyte (months).

Figure 5: Reclassification performance and early risk stratification in the eICU Collaborative Research Database. A) Reclassification Rate. Reclassification rate by analyte among Pop_{RI}-normal tests. **B) Hazard Ratios.** Cox hazard ratios for in-hospital mortality (adjusted for age and sex); analytes with $p < 0.05$ shown. **C) True Positive Rate vs False Positive Rate. D) Performance Improvement with NORMA.** Change in precision, sensitivity, specificity, and balanced accuracy (NORMA_{RI} versus Per_{RI}). **E) Lead Time.** Lead time from NORMA_{RI} flag to first Pop_{RI} flag, by analyte (hours).

2 Supplementary Figures

Figure 6: NORMA forecasting performance with Gaussian Loss. A) Forecasting Performance. Held-out test-set accuracy (mean absolute error, mean absolute percentage error, and coefficient of determination) for the Gaussian variant versus ARIMA, population mean, and last-value baselines. Error bars denote 95% bootstrap confidence intervals. **B) Per-Analyte Performance.** Per-analyte coefficient of determination grouped by clinical panel. **C) Sensitivity Analysis.** Sensitivity of Gaussian-head prediction interval width (as a percentage of Pop_{RI} width) to history length, forecast horizon, and within-person variability.

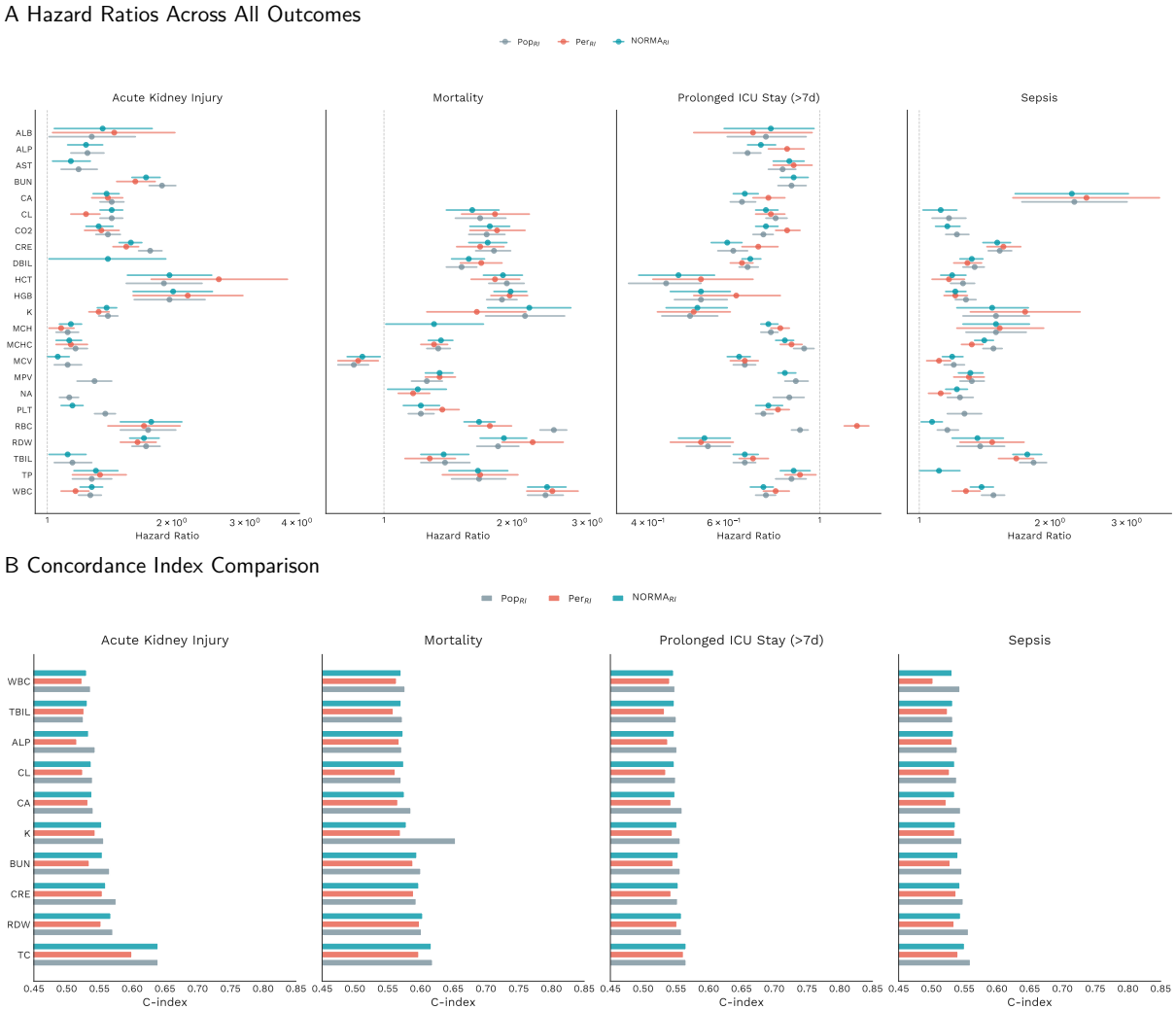

Figure 7: Proportional hazards analysis in eICU Collaborative Research Database. A) Hazard Ratios Across All Outcomes. Cox hazard ratios for in-hospital mortality, acute kidney injury, sepsis, and prolonged ICU stay, comparing Pop_{RI}, Per_{RI}, and NORMA_{RI} abnormality flags; only analytes with $p < 0.05$ are shown. **B) Concordance Index Comparison.** Concordance index for the top 10 analytes ranked by NORMA_{RI} concordance per outcome.

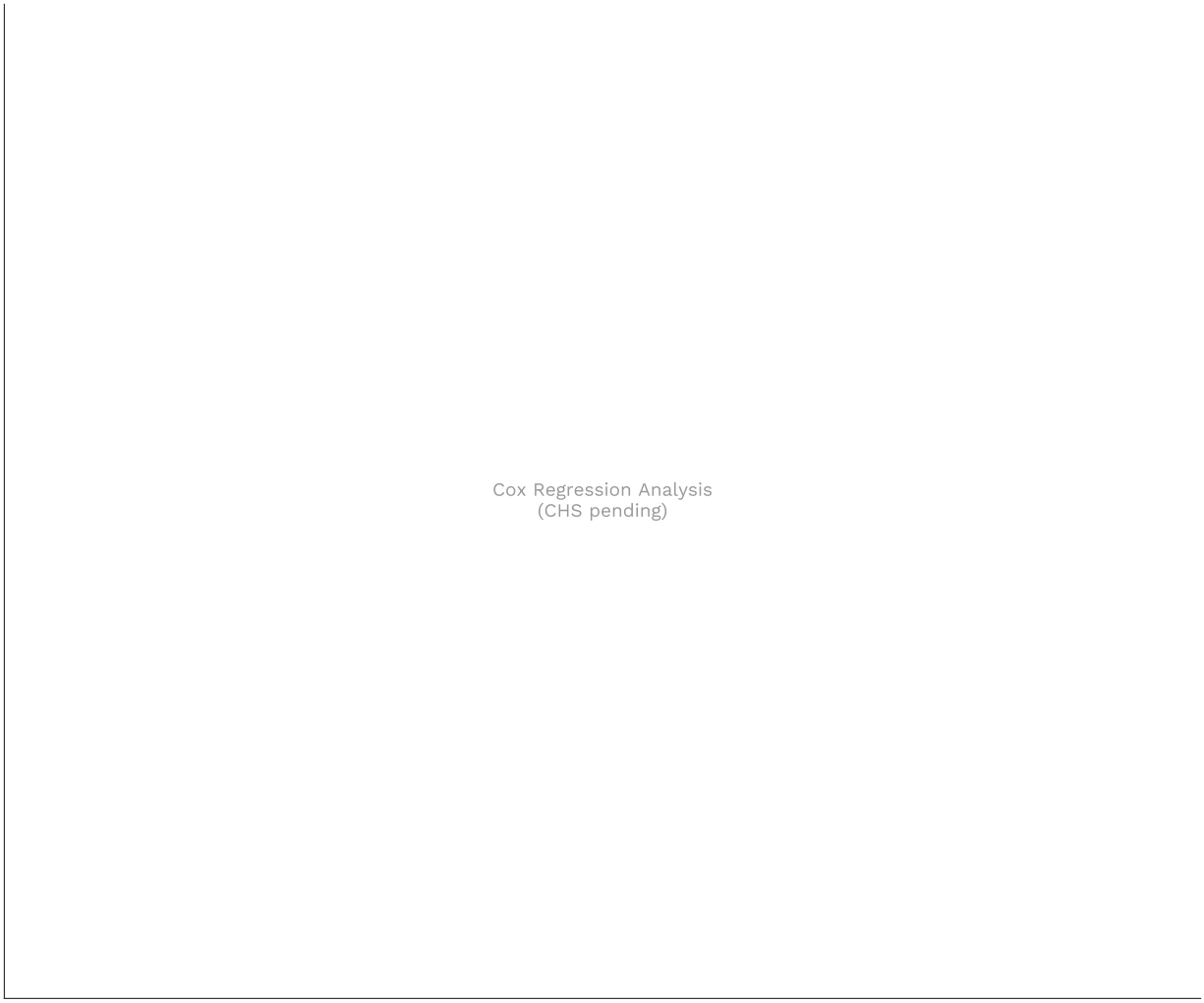

Figure 8: Proportional hazards analysis in Clalit Health Services cohort (pending).

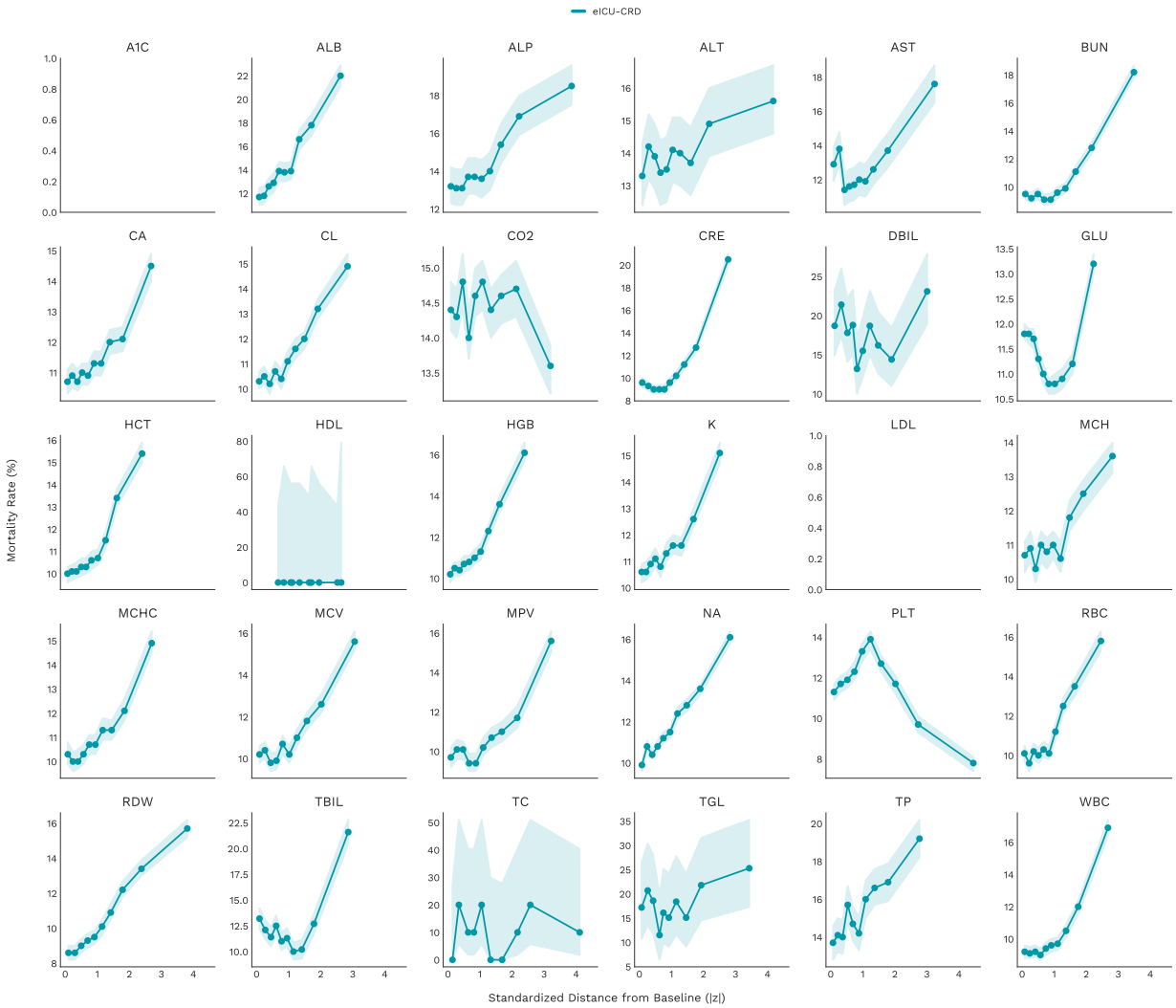

Figure 9: Deviation from the baseline associated with mortality. Each panel shows the observed mortality rate by standardized absolute deviation from the personalized baseline, binned by decile, for both Clalit Health Services and the eICU Collaborative Research Database.

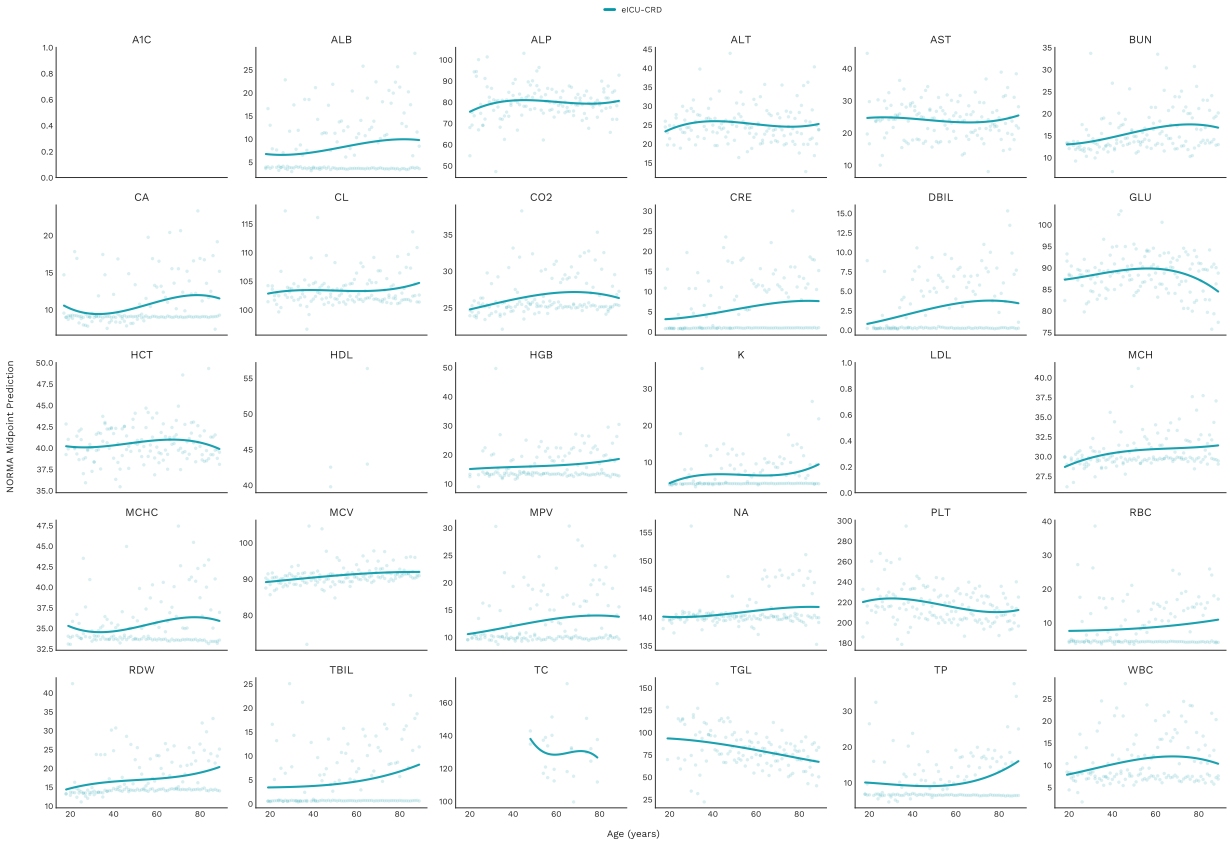

Figure 10: Age-dependent association between NORMA$_{\mathrm{RI}}$ midpoints and mortality. Each panel shows individual midpoint predictions with a polynomial smooth overlaid, for both Clalit Health Services and the eICU Collaborative Research Database.

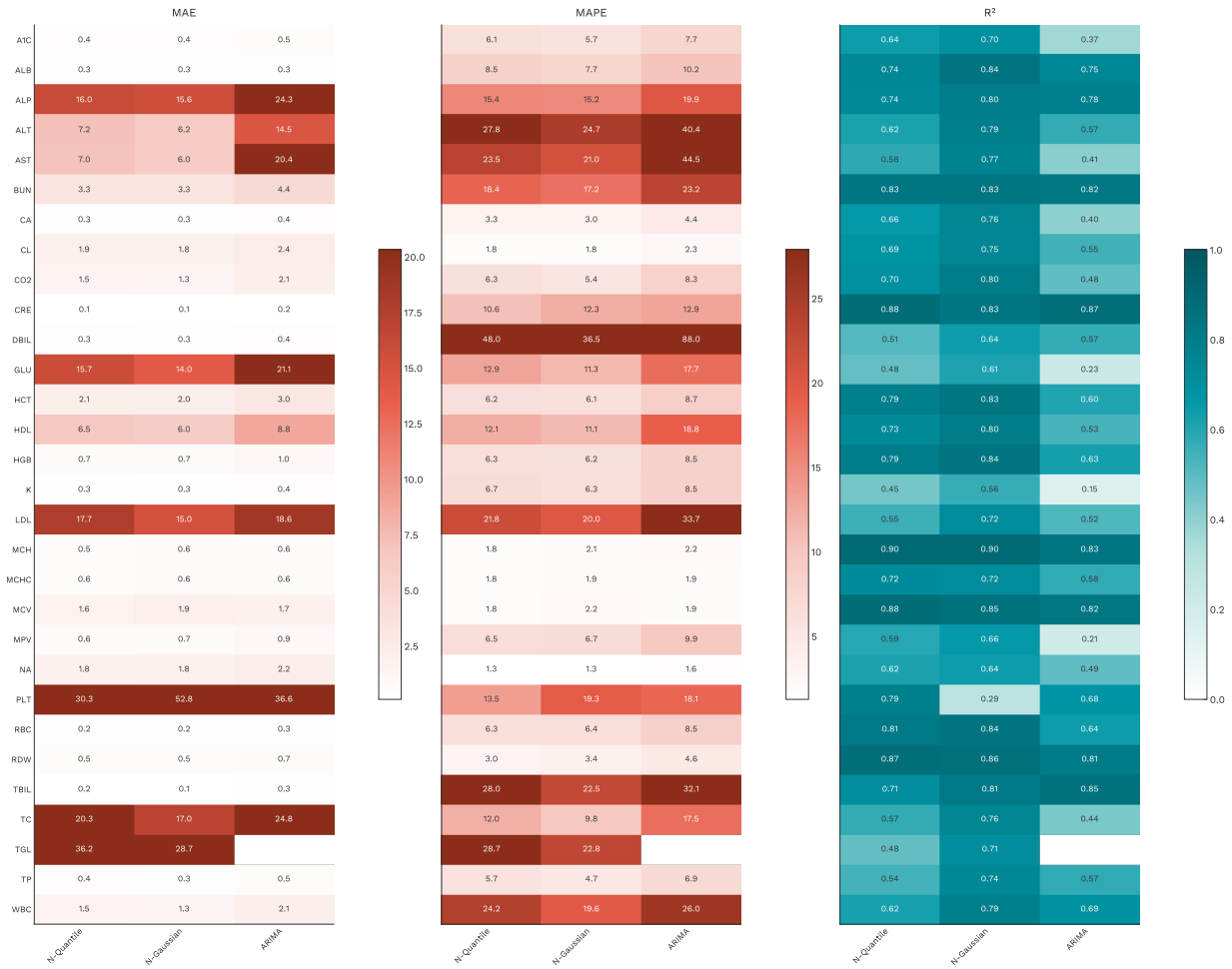

Figure 11: Per-analyte forecasting performance. NORMA using quantile and Gaussian loss, compared to ARIMA,the best performing baseline model. Color intensity encodes mean absolute error (MAE), mean absolute percentage error (MAPE), and coefficient of determination (R2) on the held-out test set.

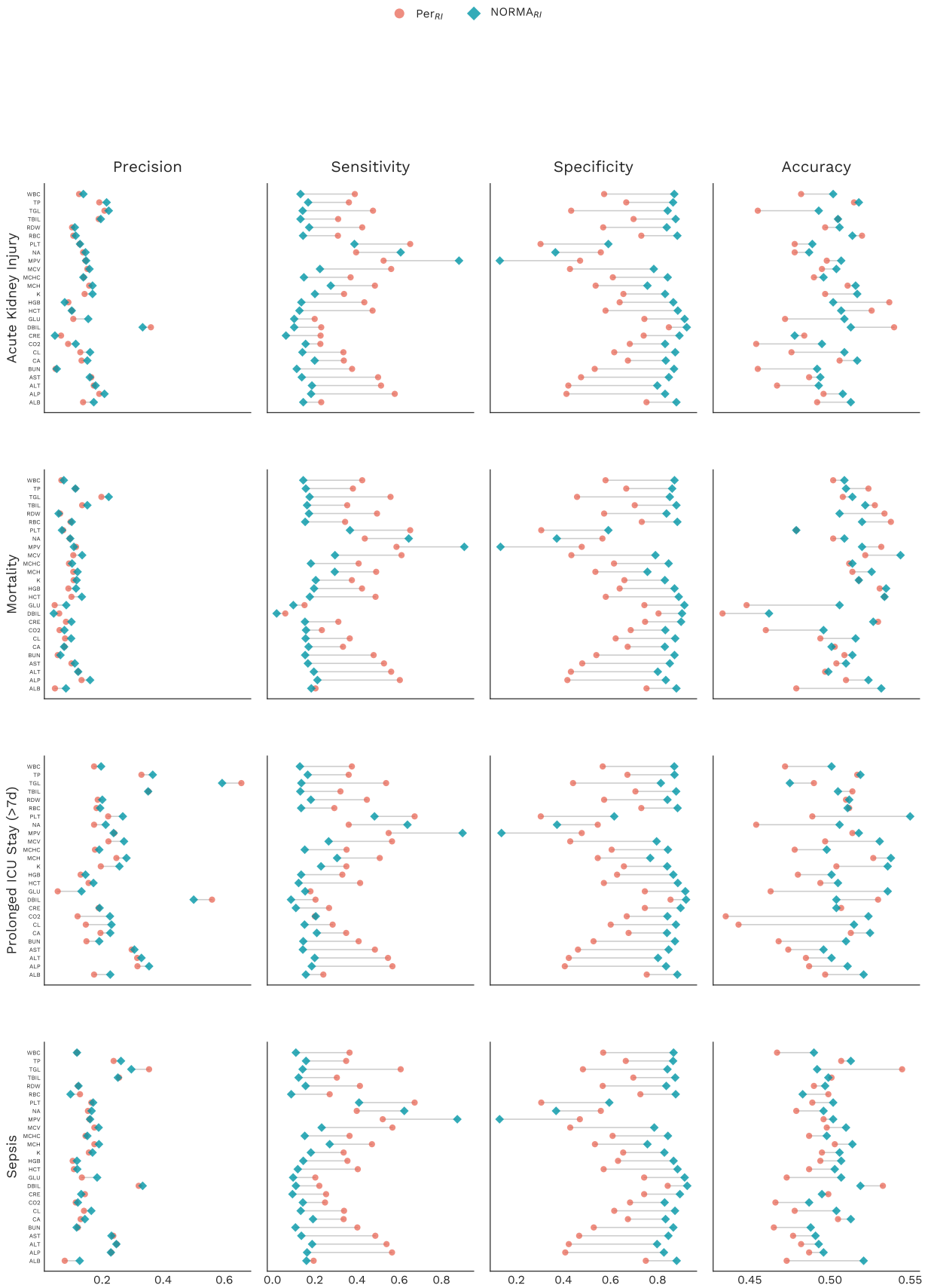

Figure 12: Clincial outcome prediction performance in the eICU Collaborative Research Database. Each point shows Per_{RI} and NORMA_{RI} values for precision, sensitivity, specificity, and balanced accuracy across outcomes. Connected pairs indicate the direction and magnitude of change between methods for each analyte. Analytes with fewer than 100 measurements are omitted.

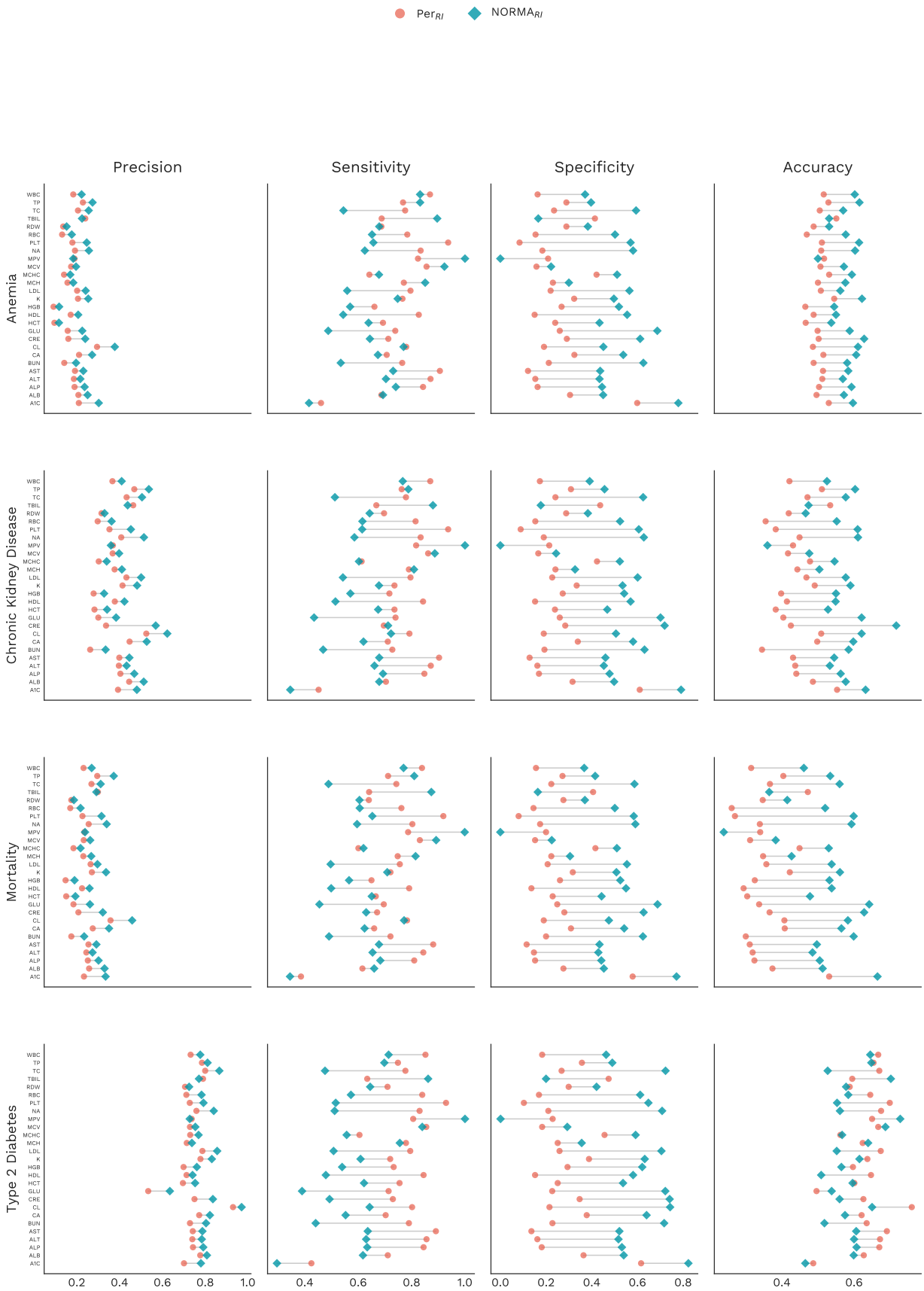

Figure 13: Clincial outcome prediction performance in the Clalit Health Services cohort. Each point shows Per_{RI} and NORMA_{RI} values for precision, sensitivity, specificity, and balanced accuracy across outcomes. Connected pairs indicate the direction and magnitude of change between methods for each analyte. Analytes with fewer than 100 measurements are omitted.

Figure 14: Abnormality prevalence by analyte and reference interval method. Grouped bars show the percentage of measurements classified as abnormal by Pop_{RI}, Per_{RI}, and NORMA_{RI}, for Clalit Health Services and the eICU Collaborative Research Database.

3 Supplementary Tables

Table 1: Analyte reference information. Full analyte name, abbreviation, unit of measurement, and conventional Pop_{RI} for all 30 laboratory analytes included in the study. Pop_{RI} were obtained from the American Board of Internal Medicine Laboratory Test Reference Ranges (January 2025).

Analyte	Full Name	Unit	Population RI
A1C	Hemoglobin A1c	%	4.0–5.6
ALB	Albumin	g/dL	3.5–5.5
ALP	Alkaline Phosphatase	U/L	30–110
ALT	Alanine Aminotransferase	U/L	10.0–40.0
AST	Aspartate Aminotransferase	U/L	10.0–40.0
BUN	Blood Urea Nitrogen	mg/dL	8–20
CA	Calcium	mg/dL	8.6–10.2
CL	Chloride	mEq/L	98–106
CO2	Bicarbonate	mEq/L	23–28
CRE	Creatinine	mg/dL	F: 0.5–1.1, M: 0.7–1.3
DBIL	Direct Bilirubin	mg/dL	0.1–0.3
GLU	Glucose	mg/dL	70–99
HCT	Hematocrit	%	F: 37–47, M: 42–50
HDL	HDL Cholesterol	mg/dL	40–100
HGB	Hemoglobin	g/dL	F: 12.0–16.0, M: 14–18
K	Potassium	mEq/L	3.5–5.0
LDL	LDL Cholesterol	mg/dL	0–130
MCH	Mean Corpuscular Hemoglobin	pg	28–32
MCHC	MCH Concentration	g/dL	33–36
MCV	Mean Corpuscular Volume	fL	80–98
MPV	Mean Platelet Volume	fL	7.5–12.5
NA	Sodium	mEq/L	136–145
PLT	Platelet Count	$10^3/\mu$L	150–450
RBC	Red Blood Cell Count	$10^6/\mu$L	F: 4.0–5.2, M: 4.5–5.9
RDW	Red Cell Distribution Width	%	9–14.5
TBIL	Total Bilirubin	mg/dL	0.3–1.1
TC	Total Cholesterol	mg/dL	100–200
TGL	Triglycerides	mg/dL	40–150
TP	Total Protein	g/dL	6.0–8.0
WBC	White Blood Cell Count	$10^3/\mu$L	4.5–11.0

Table 2: Cohort characteristics. Patient counts, demographics (age, sex, follow-up duration), and per-analyte measurement counts and mean values (plus or minus standard deviation) across all four cohorts: EHRSHOT, MIMIC-IV, eICU Collaborative Research Database, and Clalit Health Services.

	EHRSHOT		MIMIC-IV		eICU-CRD		CHS	
Patients	5,676		179,601		98,432		1,450,862	
Age	54.9 ± 17.2		58.2 ± 19.0		63.3 ± 16.1		55.1 ± 18.6	
Sex	52% F / 48% M		54% F / 46% M		46% F / 54% M		63% F / 37% M	
Time	2.5 [0.3, 7.2] yr		1.0 [0.1, 4.1] yr		8870 [5262, 15022] days		20.2 [17.5, 21.5] yr	
Analyte	N	Value	N	Value	N	Value	N	Value
A1C	1,745	6.3 ± 1.2	39,996	6.6 ± 1.3	—	—	336,493	7.1 ± 1.3
ALB	4,282	3.3 ± 0.8	55,220	3.8 ± 0.8	19,075	2.6 ± 0.7	790,253	4.1 ± 0.5
ALP	4,206	112.5 ± 58.8	78,930	107.0 ± 58.4	13,582	97.0 ± 43.2	1,099,044	82.1 ± 34.3
ALT	4,219	37.8 ± 25.2	92,599	30.2 ± 21.9	12,425	33.8 ± 23.4	1,233,902	21.9 ± 17.3
AST	4,220	31.9 ± 19.5	90,639	32.7 ± 21.4	12,704	37.1 ± 25.6	1,236,781	23.6 ± 16.3
BUN	5,180	23.0 ± 15.2	153,329	22.2 ± 14.3	57,540	25.3 ± 15.5	1,251,453	17.4 ± 8.3
CA	5,162	8.8 ± 0.7	121,478	8.8 ± 0.7	57,067	8.3 ± 0.7	909,036	9.3 ± 0.5
CL	5,172	102.4 ± 5.0	148,304	101.8 ± 5.2	59,172	103.8 ± 6.2	31,171	103.1 ± 3.9
CO2	5,187	25.9 ± 3.8	145,386	25.6 ± 4.1	70,208	25.0 ± 4.7	299	31.4 ± 12.1
CRE	5,161	1.1 ± 0.5	156,044	1.1 ± 0.6	54,820	1.1 ± 0.6	1,310,465	0.9 ± 0.4
DBIL	1,429	0.3 ± 0.3	6,635	0.7 ± 0.9	1,147	0.6 ± 0.5	—	—
GLU	5,150	124.9 ± 39.9	147,195	118.3 ± 40.4	90,958	139.6 ± 44.3	1,351,168	110.0 ± 36.1
HCT	5,470	32.2 ± 6.8	164,903	33.6 ± 6.8	60,110	30.1 ± 5.9	1,407,912	39.1 ± 5.2
HDL	1,549	53.6 ± 17.5	49,011	55.8 ± 17.7	11	26.0 ± 14.3	1,002,204	48.0 ± 15.9
HGB	5,369	10.7 ± 2.3	162,168	11.0 ± 2.3	60,482	9.9 ± 2.0	1,408,118	12.8 ± 1.8
K	5,236	4.1 ± 0.5	150,333	4.2 ± 0.5	66,569	4.0 ± 0.6	1,182,544	4.5 ± 0.5
LDL	821	94.9 ± 34.9	48,688	104.2 ± 37.8	3	92.1 ± 56.2	1,187,844	105.1 ± 33.6
MCH	5,294	30.4 ± 2.7	161,671	29.7 ± 2.8	48,432	29.7 ± 2.1	1,250,759	28.8 ± 2.3
MCHC	5,317	33.2 ± 1.4	161,806	32.6 ± 1.6	51,710	32.8 ± 1.3	1,407,568	32.8 ± 1.3
MCV	5,304	91.5 ± 6.9	161,661	91.2 ± 7.3	50,841	90.4 ± 5.7	1,400,648	87.7 ± 6.1
MPV	371	9.6 ± 1.6	—	—	37,083	9.7 ± 1.4	1,304,857	9.5 ± 1.4
NA	5,193	137.4 ± 4.2	149,303	138.7 ± 4.2	62,419	138.6 ± 5.0	1,187,537	139.9 ± 3.0
PLT	5,320	198.0 ± 111.4	162,308	229.4 ± 113.3	53,371	199.0 ± 98.4	1,407,716	243.4 ± 73.6
RBC	5,312	3.5 ± 0.8	161,825	3.7 ± 0.8	53,584	3.4 ± 0.7	1,404,794	4.5 ± 0.6
RDW	5,315	16.1 ± 2.9	161,640	15.1 ± 2.4	48,715	15.8 ± 2.0	1,403,339	14.1 ± 1.5
TBIL	4,169	0.6 ± 0.4	73,002	0.7 ± 0.6	12,738	0.8 ± 0.5	815,774	0.6 ± 0.4
TC	1,635	173.5 ± 45.0	49,762	187.5 ± 44.6	35	108.3 ± 44.0	1,260,730	181.7 ± 40.7
TGL	4	165.7 ± 100.8	48,445	140.4 ± 77.5	312	144.9 ± 68.0	1,244,518	133.5 ± 65.3
TP	4,213	6.6 ± 1.0	18,673	7.0 ± 0.7	15,134	5.8 ± 1.0	791,481	7.1 ± 0.6
WBC	5,345	7.8 ± 4.6	164,291	7.7 ± 4.0	52,141	10.6 ± 4.6	1,409,027	7.5 ± 3.7

Table 3: Within-person and between-person biological variability. Per-analyte within-person coefficient of variation, between-person coefficient of variation, and individuality index for the eICU Collaborative Research Database and Clalit Health Services cohorts. Values are reported as median [95% confidence interval].

	eICU-CRD			CHS		
Analyte	CV_{intra}	CV_{inter}	II	CV_{intra}	CV_{inter}	II
A1C	—	—	—	8.6 [8.5, 8.6]	16.5 [16.5, 16.6]	0.5 [0.5, 0.5]
ALB	12.8 [12.7, 12.9]	21.7 [21.5, 21.9]	0.6 [0.6, 0.6]	6.7 [6.7, 6.8]	18.9 [18.3, 19.4]	0.4 [0.3, 0.4]
ALP	17.7 [17.6, 17.9]	40.4 [40.1, 40.7]	0.4 [0.4, 0.4]	17.8 [17.7, 17.8]	30.1 [30.0, 30.2]	0.6 [0.6, 0.6]
ALT	26.5 [26.2, 26.7]	61.6 [61.2, 62.0]	0.4 [0.4, 0.4]	34.2 [34.2, 34.3]	50.2 [50.0, 50.5]	0.7 [0.7, 0.7]
AST	31.9 [31.6, 32.1]	58.6 [58.2, 59.0]	0.5 [0.5, 0.6]	24.7 [24.6, 24.7]	35.1 [34.9, 35.4]	0.7 [0.7, 0.7]
BUN	26.3 [26.2, 26.4]	55.2 [55.0, 55.4]	0.5 [0.5, 0.5]	20.5 [20.4, 20.5]	36.2 [36.2, 36.3]	0.6 [0.6, 0.6]
CA	5.3 [5.2, 5.3]	6.9 [6.8, 6.9]	0.8 [0.8, 0.8]	3.5 [3.5, 3.5]	3.7 [3.7, 3.7]	0.9 [0.9, 0.9]
CL	3.2 [3.2, 3.2]	4.8 [4.7, 4.8]	0.7 [0.7, 0.7]	2.4 [2.3, 2.4]	2.3 [2.2, 2.3]	1.0 [1.0, 1.1]
CO2	10.6 [10.5, 10.6]	15.1 [15.1, 15.2]	0.7 [0.7, 0.7]	17.7 [17.3, 18.2]	27.0 [25.9, 28.1]	0.7 [0.6, 0.7]
CRE	18.1 [18.0, 18.1]	46.3 [46.1, 46.5]	0.4 [0.4, 0.4]	12.2 [12.2, 12.2]	34.7 [34.4, 34.9]	0.3 [0.3, 0.4]
DBIL	36.8 [35.5, 38.1]	79.9 [78.1, 81.7]	0.5 [0.4, 0.5]	—	—	—
GLU	21.4 [21.4, 21.5]	22.2 [22.1, 22.3]	1.0 [1.0, 1.0]	13.7 [13.7, 13.7]	24.0 [23.9, 24.0]	0.6 [0.6, 0.6]
HCT	10.3 [10.3, 10.3]	17.6 [17.5, 17.7]	0.6 [0.6, 0.6]	6.3 [6.3, 6.4]	9.6 [9.6, 9.6]	0.7 [0.7, 0.7]
HDL	11.6 [9.3, 14.4]	39.7 [34.0, 44.9]	0.3 [0.2, 0.4]	24.4 [24.4, 24.5]	30.1 [30.1, 30.2]	0.8 [0.8, 0.8]
HGB	10.4 [10.4, 10.5]	18.4 [18.3, 18.5]	0.6 [0.6, 0.6]	6.3 [6.3, 6.3]	10.5 [10.5, 10.5]	0.6 [0.6, 0.6]
K	10.0 [9.9, 10.0]	9.9 [9.9, 10.0]	1.0 [1.0, 1.0]	7.1 [7.1, 7.1]	6.5 [6.4, 6.5]	1.1 [1.1, 1.1]
LDL	13.6 [9.1, 19.3]	44.5 [37.3, 50.3]	0.3 [0.2, 0.4]	20.6 [20.5, 20.6]	23.7 [23.7, 23.8]	0.9 [0.9, 0.9]
MCH	1.5 [1.5, 1.5]	6.9 [6.9, 6.9]	0.2 [0.2, 0.2]	3.6 [3.6, 3.6]	7.1 [7.1, 7.1]	0.5 [0.5, 0.5]
MCHC	1.8 [1.8, 1.8]	3.6 [3.5, 3.6]	0.5 [0.5, 0.5]	2.9 [2.9, 2.9]	2.7 [2.7, 2.7]	1.1 [1.1, 1.1]
MCV	1.6 [1.5, 1.6]	6.2 [6.1, 6.2]	0.2 [0.2, 0.2]	3.2 [3.2, 3.2]	6.1 [6.1, 6.1]	0.5 [0.5, 0.5]
MPV	3.8 [3.8, 3.8]	12.7 [12.7, 12.8]	0.3 [0.3, 0.3]	10.3 [10.3, 10.3]	10.8 [10.8, 10.8]	0.9 [0.9, 1.0]
NA	1.9 [1.9, 1.9]	2.7 [2.7, 2.7]	0.7 [0.7, 0.7]	1.5 [1.5, 1.5]	1.2 [1.2, 1.2]	1.2 [1.2, 1.2]
PLT	20.0 [19.9, 20.1]	39.3 [39.1, 39.5]	0.5 [0.5, 0.5]	14.1 [14.1, 14.1]	23.8 [23.7, 23.8]	0.6 [0.6, 0.6]
RBC	10.1 [10.1, 10.2]	18.0 [18.0, 18.1]	0.6 [0.6, 0.6]	5.9 [5.9, 5.9]	10.1 [10.1, 10.1]	0.6 [0.6, 0.6]
RDW	2.9 [2.9, 3.0]	12.5 [12.4, 12.6]	0.2 [0.2, 0.2]	5.7 [5.7, 5.7]	7.1 [7.1, 7.1]	0.8 [0.8, 0.8]
TBIL	29.0 [28.8, 29.2]	56.5 [56.1, 56.9]	0.5 [0.5, 0.5]	26.0 [25.9, 26.0]	42.8 [42.6, 43.0]	0.6 [0.6, 0.6]
TC	11.5 [9.9, 13.3]	38.0 [34.5, 41.0]	0.3 [0.3, 0.4]	13.9 [13.9, 14.0]	16.6 [16.6, 16.6]	0.8 [0.8, 0.8]
TGL	25.1 [24.2, 26.0]	41.8 [40.5, 43.2]	0.6 [0.6, 0.6]	26.1 [26.1, 26.1]	40.3 [40.2, 40.4]	0.7 [0.7, 0.7]
TP	9.7 [9.6, 9.8]	13.7 [13.6, 13.8]	0.7 [0.7, 0.7]	4.8 [4.8, 4.8]	5.6 [5.5, 5.6]	0.9 [0.9, 0.9]
WBC	25.1 [25.0, 25.2]	34.4 [34.2, 34.5]	0.7 [0.7, 0.7]	19.2 [19.2, 19.3]	51.6 [28.8, 79.2]	0.4 [0.2, 0.7]

Table 4: NORMA forecasting performance. Mean absolute error (MAE), mean absolute percentage error (MAPE), and coefficient of determination (R2) for NORMA Gaussian, NORMA quantile, ARIMA, population mean, and last-value-carried-forward, reported on training, validation, and held-out test sets. Values are reported as median [interquartile range] across analytes.

Metric		NORMA Gaussian	NORMA Quantile	ARIMA	Mean	Last
MAE	Train	6.0 [3.2, 11.1]	**5.7 [2.9, 10.1]**	6.8 [2.5, 10.2]	8.4 [4.5, 15.0]	7.4 [3.7, 11.0]
	Val	6.0 [2.6, 10.7]	**5.8 [2.9, 8.7]**	6.8 [3.6, 10.0]	8.3 [4.9, 13.5]	7.3 [3.8, 10.3]
	Test	6.0 [2.6, 10.2]	**5.9 [3.6, 9.1]**	6.7 [4.3, 11.9]	8.4 [4.5, 13.6]	7.3 [3.9, 10.7]
MAPE	Train	**10.9 [8.8, 14.0]**	11.9 [8.6, 14.7]	15.9 [11.3, 20.1]	19.4 [14.8, 25.4]	14.8 [10.9, 19.6]
	Val	**11.1 [7.9, 13.2]**	12.6 [8.7, 15.6]	15.7 [12.1, 21.5]	19.7 [14.0, 27.4]	15.2 [10.6, 20.1]
	Test	**11.1 [8.5, 14.3]**	12.3 [7.9, 17.3]	16.9 [11.3, 23.3]	19.7 [13.5, 26.7]	15.2 [10.6, 21.1]
R2	Train	**0.76 [0.72, 0.80]**	0.70 [0.66, 0.75]	0.62 [0.57, 0.70]	0.46 [0.41, 0.49]	0.56 [0.47, 0.62]
	Val	**0.75 [0.69, 0.78]**	0.69 [0.66, 0.73]	0.62 [0.53, 0.67]	0.46 [0.43, 0.49]	0.55 [0.49, 0.62]
	Test	**0.75 [0.70, 0.78]**	0.68 [0.64, 0.72]	0.58 [0.51, 0.64]	0.45 [0.41, 0.50]	0.54 [0.49, 0.61]

Table 5: Sensitivity of NORMA prediction intervals to patient features. Median and interquartile range of prediction interval width change (as a percentage of Pop_{RI} width) and midpoint shift (as a percentage of Pop_{RI} width) in response to history length, forecast horizon, and within-person variability, for both the quantile and Gaussian parameterizations.

		CI Width Change (%)		Midpoint Shift (%)	
Model	Feature	Median	IQR	Median	IQR
Quantile	History Length	4.8	[1.0, 14.4]	0.0	[0.0, 0.0]
Quantile	Prediction Horizon	1.6	[0.1, 11.1]	0.0	[0.0, 0.0]
Quantile	Within-Person Variability	115.2	[111.4, 120.5]	0.7	[0.3, 1.7]
Gaussian	History Length	28.3	[20.6, 41.8]	12.8	[5.6, 20.6]
Gaussian	Prediction Horizon	7.2	[3.0, 9.9]	1.6	[0.8, 3.2]
Gaussian	Within-Person Variability	7.2	[3.7, 12.4]	1.3	[0.3, 3.6]

Table 6: Abnormality prevalence and reclassification rates. Overall percentage of measurements classified as abnormal by Pop_{RI}, Per_{RI}, and NORMA_{RI}, and the reclassification rate (the proportion of Pop_{RI}-normal tests flagged abnormal by Per_{RI} or NORMA_{RI}) in the eICU Collaborative Research Database and Clalit Health Services.

Dataset	N	Pop_{RI} (%)	Per_{RI} (%)	NORMA_{RI} (%)	Per_{RI} RR (%)	NORMA_{RI} RR (%)
eICU-CRD	29	50.2	68.1	55.8	37.3	9.2
CHS	28	18.9	38.9	29.2	24.3	12.0

Table 7: Per-analyte reclassification. Among tests with a Per_{RI} setpoint within Pop_{RI}, the number per 1,000 flagged abnormal by Pop_{RI}, Per_{RI}, and NORMA_{RI} for each analyte, in the eICU Collaborative Research Database and Clalit Health Services.

	CHS			eICU-CRD		
Analyte	Pop_{RI}	Per_{RI}	NORMA_{RI}	Pop_{RI}	Per_{RI}	NORMA_{RI}
A1C	240	315	282	—	—	—
ALB	146	284	244	466	551	504
ALP	119	435	218	224	535	294
ALT	136	414	210	252	531	327
AST	71	383	142	186	441	240
BUN	235	458	267	360	546	400
CA	140	290	199	245	378	300
CL	229	415	282	322	461	360
CO2	545	602	562	428	512	467
CRE	141	256	191	231	333	266
DBIL	—	—	—	136	215	172
GLU	326	517	350	541	586	556
HCT	322	497	377	535	677	566
HDL	157	445	217	—	1,000	—
HGB	279	424	331	518	643	555
K	122	250	186	148	281	200
LDL	118	414	195	—	800	—
MCH	184	363	290	83	308	177
MCHC	400	498	439	289	452	339
MCV	94	375	266	48	372	141
MPV	76	314	880	44	320	616
NA	111	361	150	246	418	300
PLT	75	412	125	122	524	310
RBC	240	486	291	475	570	509
RDW	234	378	304	248	456	312
TBIL	119	220	489	149	293	200
TC	184	443	240	286	743	400
TGL	—	—	—	212	519	285
TP	102	273	220	277	427	329
WBC	146	377	228	256	435	298

Table 8: In-hospital mortality prediction performance in the eICU Collaborative Research Database. Per-analyte precision, sensitivity, specificity, and balanced accuracy for Per_{RI} and NORMA_{RI} abnormality flags, restricted to measurements classified as normal by Pop_{RI}. Analytes with fewer than 100 measurements are omitted (—).

			Per_{RI}				NORMA_{RI}			
Analyte	N	Events	Precision	Sensitivity	Specificity	Accuracy	Precision	Sensitivity	Specificity	Accuracy
A1C	—	—	—	—	—	—	—	—	—	—
ALB	891	49	0.05	0.20	0.75	0.48	0.08	0.18	0.88	0.53
ALP	8,843	1,143	0.13	0.60	0.42	0.51	0.16	0.21	0.83	0.52
ALT	7,391	913	0.12	0.56	0.43	0.50	0.12	0.20	0.80	0.50
AST	8,035	793	0.10	0.53	0.48	0.50	0.11	0.17	0.85	0.51
BUN	19,881	1,058	0.06	0.48	0.54	0.51	0.06	0.15	0.87	0.51
CA	16,083	1,214	0.08	0.33	0.67	0.50	0.08	0.17	0.83	0.50
CL	26,608	2,149	0.08	0.37	0.62	0.49	0.10	0.16	0.88	0.52
CO2	29,891	2,393	0.06	0.23	0.69	0.46	0.08	0.16	0.83	0.50
CRE	27,847	1,883	0.08	0.31	0.75	0.53	0.10	0.15	0.90	0.53
DBIL	289	49	0.06	0.06	0.80	0.43	0.04	0.02	0.90	0.46
GLU	9,534	696	0.04	0.15	0.74	0.45	0.08	0.10	0.91	0.51
HCT	2,919	254	0.10	0.49	0.58	0.53	0.13	0.18	0.89	0.53
HDL	—	—	—	—	—	—	—	—	—	—
HGB	3,383	264	0.09	0.42	0.64	0.53	0.12	0.20	0.87	0.54
K	57,801	5,680	0.11	0.38	0.66	0.52	0.12	0.20	0.83	0.52
LDL	—	—	—	—	—	—	—	—	—	—
MCH	32,416	3,272	0.11	0.49	0.54	0.51	0.12	0.29	0.76	0.53
MCHC	22,846	1,998	0.09	0.41	0.61	0.51	0.10	0.18	0.85	0.51
MCV	43,792	4,335	0.11	0.61	0.43	0.52	0.14	0.30	0.79	0.54
MPV	34,479	3,564	0.12	0.59	0.48	0.53	0.11	0.91	0.13	0.52
NA	40,769	3,828	0.10	0.44	0.57	0.50	0.10	0.65	0.37	0.51
PLT	33,546	2,580	0.07	0.65	0.31	0.48	0.07	0.37	0.59	0.48
RBC	4,595	355	0.10	0.34	0.73	0.54	0.10	0.15	0.88	0.52
RDW	16,275	889	0.06	0.49	0.57	0.53	0.06	0.17	0.84	0.51
TBIL	8,985	1,041	0.14	0.35	0.70	0.53	0.15	0.16	0.88	0.52
TC	11	2	0.29	1.00	0.44	0.72	0.67	1.00	0.89	0.94
TGL	176	34	0.20	0.56	0.46	0.51	0.22	0.18	0.85	0.51
TP	4,859	486	0.11	0.38	0.67	0.52	0.11	0.16	0.86	0.51
WBC	25,827	1,714	0.07	0.42	0.58	0.50	0.07	0.15	0.87	0.51

Table 9: Acute kidney injury prediction performance in the eICU Collaborative Research Database. Per-analyte precision, sensitivity, specificity, and balanced accuracy for Per_{RI} and NORMA_{RI} abnormality flags, restricted to measurements classified as normal by Pop_{RI}. Analytes with fewer than 100 measurements are omitted (—).

			Per_{RI}				NORMA_{RI}			
Analyte	N	Events	Precision	Sensitivity	Specificity	Accuracy	Precision	Sensitivity	Specificity	Accuracy
A1C	—	—	—	—	—	—	—	—	—	—
ALB	891	130	0.14	0.23	0.75	0.49	0.17	0.15	0.88	0.51
ALP	8,843	1,712	0.19	0.58	0.41	0.50	0.21	0.18	0.83	0.51
ALT	7,391	1,408	0.17	0.51	0.42	0.47	0.18	0.19	0.80	0.49
AST	8,035	1,384	0.17	0.50	0.47	0.49	0.16	0.14	0.85	0.49
BUN	19,881	1,164	0.05	0.38	0.53	0.46	0.05	0.12	0.87	0.49
CA	16,083	2,079	0.13	0.34	0.67	0.51	0.15	0.20	0.83	0.52
CL	26,608	3,839	0.13	0.34	0.62	0.48	0.16	0.14	0.88	0.51
CO2	29,891	3,601	0.09	0.23	0.68	0.45	0.11	0.16	0.83	0.49
CRE	27,847	2,062	0.07	0.23	0.74	0.48	0.05	0.06	0.89	0.48
DBIL	289	78	0.36	0.23	0.85	0.54	0.33	0.10	0.92	0.51
GLU	9,534	1,257	0.11	0.20	0.74	0.47	0.15	0.10	0.92	0.51
HCT	2,919	266	0.10	0.47	0.58	0.53	0.10	0.13	0.89	0.51
HDL	—	—	—	—	—	—	—	—	—	—
HGB	3,383	255	0.09	0.43	0.64	0.54	0.08	0.14	0.87	0.50
K	57,801	8,390	0.14	0.34	0.66	0.50	0.17	0.20	0.83	0.52
LDL	—	—	—	—	—	—	—	—	—	—
MCH	32,416	4,932	0.16	0.48	0.54	0.51	0.17	0.28	0.76	0.52
MCHC	22,846	3,339	0.14	0.37	0.61	0.49	0.14	0.15	0.84	0.50
MCV	43,792	6,744	0.15	0.56	0.43	0.49	0.16	0.23	0.78	0.50
MPV	34,479	5,045	0.15	0.53	0.47	0.50	0.15	0.88	0.13	0.51
NA	40,769	6,154	0.14	0.40	0.56	0.48	0.15	0.61	0.37	0.49
PLT	33,546	4,508	0.13	0.65	0.30	0.48	0.13	0.39	0.59	0.49
RBC	4,595	429	0.11	0.31	0.73	0.52	0.11	0.14	0.88	0.51
RDW	16,275	1,681	0.10	0.42	0.57	0.50	0.11	0.17	0.84	0.51
TBIL	8,985	1,660	0.19	0.31	0.70	0.51	0.20	0.13	0.88	0.51
TC	11	3	0.29	0.67	0.38	0.52	0.33	0.33	0.75	0.54
TGL	176	42	0.21	0.48	0.43	0.46	0.22	0.14	0.84	0.49
TP	4,859	864	0.19	0.36	0.67	0.52	0.21	0.17	0.87	0.52
WBC	25,827	3,491	0.12	0.39	0.57	0.48	0.14	0.13	0.87	0.50

Table 10: Sepsis prediction performance in the eICU Collaborative Research Database. Per-analyte precision, sensitivity, specificity, and balanced accuracy for Per_{RI} and NORMA_{RI} abnormality flags, restricted to measurements classified as normal by Pop_{RI}. Analytes with fewer than 100 measurements are omitted (—).

			Per_{RI}				NORMA_{RI}			
Analyte	N	Events	Precision	Sensitivity	Specificity	Accuracy	Precision	Sensitivity	Specificity	Accuracy
A1C	—	—	—	—	—	—	—	—	—	—
ALB	891	87	0.08	0.20	0.75	0.47	0.13	0.16	0.88	0.52
ALP	8,843	2,106	0.23	0.57	0.41	0.49	0.23	0.16	0.83	0.50
ALT	7,391	1,925	0.25	0.54	0.42	0.48	0.25	0.19	0.80	0.49
AST	8,035	2,036	0.24	0.49	0.47	0.48	0.23	0.14	0.85	0.49
BUN	19,881	2,759	0.12	0.40	0.53	0.47	0.12	0.11	0.87	0.49
CA	16,083	2,034	0.13	0.34	0.67	0.51	0.14	0.19	0.83	0.51
CL	26,608	4,164	0.14	0.34	0.62	0.48	0.17	0.13	0.87	0.50
CO2	29,891	4,190	0.11	0.25	0.68	0.47	0.12	0.14	0.83	0.49
CRE	27,847	4,045	0.14	0.25	0.74	0.50	0.13	0.10	0.90	0.49
DBIL	289	72	0.32	0.22	0.84	0.53	0.33	0.11	0.93	0.52
GLU	9,534	1,557	0.13	0.20	0.74	0.47	0.18	0.10	0.92	0.51
HCT	2,919	332	0.11	0.40	0.57	0.49	0.12	0.12	0.89	0.50
HDL	—	—	—	—	—	—	—	—	—	—
HGB	3,383	363	0.10	0.35	0.63	0.49	0.12	0.15	0.87	0.51
K	57,801	9,268	0.16	0.34	0.65	0.49	0.17	0.18	0.83	0.51
LDL	—	—	—	—	—	—	—	—	—	—
MCH	32,416	5,623	0.17	0.47	0.53	0.50	0.19	0.27	0.76	0.51
MCHC	22,846	3,540	0.15	0.36	0.61	0.49	0.15	0.15	0.84	0.50
MCV	43,792	7,707	0.17	0.57	0.43	0.50	0.19	0.23	0.79	0.51
MPV	34,479	5,542	0.16	0.52	0.47	0.50	0.16	0.88	0.13	0.50
NA	40,769	6,847	0.15	0.40	0.56	0.48	0.17	0.62	0.37	0.50
PLT	33,546	5,676	0.17	0.67	0.31	0.49	0.17	0.41	0.59	0.50
RBC	4,595	594	0.13	0.27	0.73	0.50	0.10	0.09	0.88	0.48
RDW	16,275	2,064	0.12	0.41	0.57	0.49	0.12	0.16	0.84	0.50
TBIL	8,985	2,286	0.26	0.30	0.70	0.50	0.25	0.12	0.88	0.50
TC	11	2	0.29	1.00	0.44	0.72	0.00	0.00	0.67	0.33
TGL	176	56	0.35	0.61	0.48	0.55	0.30	0.14	0.84	0.49
TP	4,859	1,118	0.24	0.35	0.67	0.51	0.26	0.16	0.87	0.51
WBC	25,827	3,531	0.12	0.36	0.57	0.47	0.12	0.11	0.87	0.49

Table 11: Prolonged ICU stay prediction performance in the eICU Collaborative Research Database. Per-analyte precision, sensitivity, specificity, and balanced accuracy for Per_{RI} and NORMA_{RI} abnormality flags, restricted to measurements classified as normal by Pop_{RI}. Analytes with fewer than 100 measurements are omitted (—).

			Per_{RI}				NORMA_{RI}			
Analyte	N	Events	Precision	Sensitivity	Specificity	Accuracy	Precision	Sensitivity	Specificity	Accuracy
A1C	—	—	—	—	—	—	—	—	—	—
ALB	891	158	0.17	0.24	0.75	0.50	0.23	0.16	0.88	0.52
ALP	8,843	2,882	0.32	0.57	0.41	0.49	0.35	0.19	0.84	0.51
ALT	7,391	2,412	0.32	0.55	0.42	0.48	0.33	0.20	0.80	0.50
AST	8,035	2,561	0.30	0.48	0.46	0.47	0.31	0.14	0.85	0.50
BUN	19,881	3,352	0.15	0.41	0.53	0.47	0.19	0.15	0.87	0.51
CA	16,083	2,941	0.20	0.35	0.68	0.51	0.23	0.21	0.84	0.53
CL	26,608	5,167	0.15	0.28	0.60	0.44	0.23	0.15	0.88	0.52
CO2	29,891	5,495	0.12	0.20	0.67	0.43	0.23	0.20	0.84	0.52
CRE	27,847	5,026	0.19	0.27	0.75	0.51	0.19	0.11	0.90	0.50
DBIL	289	137	0.56	0.20	0.85	0.53	0.50	0.09	0.92	0.50
GLU	9,534	718	0.06	0.18	0.75	0.46	0.13	0.15	0.92	0.54
HCT	2,919	467	0.16	0.41	0.57	0.49	0.17	0.12	0.89	0.51
HDL	—	—	—	—	—	—	—	—	—	—
HGB	3,383	486	0.13	0.33	0.63	0.48	0.15	0.14	0.87	0.50
K	57,801	11,154	0.20	0.35	0.66	0.50	0.26	0.23	0.84	0.54
LDL	—	—	—	—	—	—	—	—	—	—
MCH	32,416	7,342	0.25	0.51	0.55	0.53	0.28	0.31	0.77	0.54
MCHC	22,846	4,454	0.18	0.35	0.60	0.48	0.19	0.15	0.84	0.50
MCV	43,792	9,765	0.22	0.57	0.43	0.50	0.27	0.27	0.80	0.53
MPV	34,479	7,956	0.24	0.55	0.48	0.51	0.24	0.90	0.14	0.52
NA	40,769	8,509	0.17	0.36	0.55	0.45	0.21	0.64	0.37	0.51
PLT	33,546	7,568	0.22	0.67	0.30	0.49	0.27	0.48	0.62	0.55
RBC	4,595	778	0.18	0.29	0.73	0.51	0.19	0.14	0.89	0.51
RDW	16,275	2,917	0.19	0.45	0.57	0.51	0.20	0.18	0.84	0.51
TBIL	8,985	2,983	0.35	0.32	0.71	0.51	0.35	0.13	0.88	0.51
TC	11	1	0.00	0.00	0.30	0.15	0.33	1.00	0.80	0.90
TGL	176	117	0.66	0.54	0.44	0.49	0.59	0.14	0.81	0.47
TP	4,859	1,495	0.33	0.36	0.67	0.52	0.37	0.17	0.87	0.52
WBC	25,827	5,040	0.17	0.38	0.57	0.47	0.20	0.13	0.87	0.50

Table 12: Positive predictive value across outcomes in the eICU Collaborative Research Database. Measurements classified as normal by Pop_{RI} are excluded. Per 100 patients flagged abnormal by Per_{RI} or NORMA_{RI}, the number who experienced each outcome (acute kidney injury, in-hospital mortality, prolonged ICU stay, and sepsis), by analyte. Analytes with fewer than 100 measurements are omitted (—).

	Acute Kidney Injury		Mortality		Prolonged ICU Stay (>7d)		Sepsis	
Analyte	Per_{RI}	NORMA_{RI}	Per_{RI}	NORMA_{RI}	Per_{RI}	NORMA_{RI}	Per_{RI}	NORMA_{RI}
A1C	—	—	—	—	—	—	—	—
ALB	14	17	5	8	17	23	8	13
ALP	19	21	13	16	32	35	23	23
ALT	17	18	12	12	32	33	25	25
AST	16	16	10	11	30	31	24	23
BUN	5	5	6	6	15	19	12	12
CA	13	15	8	8	20	23	13	14
CL	13	16	8	10	15	23	14	16
CO2	9	11	6	8	12	23	11	12
CRE	7	5	8	10	19	19	14	13
DBIL	36	33	6	4	56	50	32	33
GLU	11	16	4	8	6	13	13	18
HCT	10	10	10	13	16	17	11	12
HDL	—	—	—	—	—	—	—	—
HGB	9	8	9	12	13	15	10	12
K	14	17	11	12	20	26	16	17
LDL	—	—	—	—	—	—	—	—
MCH	16	17	11	12	25	28	18	19
MCHC	14	14	9	10	18	19	15	15
MCV	15	16	11	14	22	27	18	19
MPV	15	15	12	11	24	24	16	16
NA	14	15	10	10	17	21	15	17
PLT	13	13	7	7	22	27	16	17
RBC	11	11	10	10	18	19	13	10
RDW	10	11	6	6	19	20	12	12
TBIL	19	20	14	15	35	35	26	25
TC	29	33	29	67	0	33	29	0
TGL	21	22	20	22	66	59	35	30
TP	19	22	11	11	33	37	24	26
WBC	12	14	7	8	17	20	12	12
Overall	**15**	**16**	**10**	**13**	**23**	**27**	**18**	**17**

Table 13: Positive predictive value across outcomes in Clalit Health Services cohort. Measurements classified as normal by Pop_{RI} are excluded. Per 100 patients flagged abnormal by Per_{RI} or NORMA_{RI}, the number who experienced each outcome (chronic kidney disease, all-cause mortality, and type 2 diabetes), by analyte. Analytes with fewer than 100 measurements are omitted (—).

	Anemia		Chronic Kidney Disease		Mortality		Type 2 Diabetes	
Analyte	Per_{RI}	NORMA_{RI}	Per_{RI}	NORMA_{RI}	Per_{RI}	NORMA_{RI}	Per_{RI}	NORMA_{RI}
A1C	21	30	39	48	23	34	70	78
ALB	21	25	44	51	26	33	78	81
ALP	19	24	40	47	25	30	74	79
ALT	19	22	40	43	24	27	74	78
AST	19	23	40	45	26	29	74	79
BUN	14	20	26	34	18	24	73	80
CA	21	27	45	53	28	35	77	82
CL	30	38	52	62	36	46	93	97
CO2	100	100	—	—	—	—	—	—
CRE	16	24	34	57	21	32	75	84
DBIL	—	—	—	—	—	—	—	—
GLU	16	23	30	38	18	26	53	64
HCT	10	12	28	34	15	19	70	75
HDL	17	21	38	42	22	26	71	74
HGB	9	12	28	33	15	19	70	76
K	21	25	41	48	27	34	78	83
LDL	20	24	43	50	26	30	79	86
MCH	16	18	38	41	23	27	71	74
MCHC	14	17	30	34	18	22	73	77
MCV	17	20	37	40	23	26	73	75
MPV	19	18	37	36	24	24	74	73
NA	19	26	41	51	26	34	76	84
PLT	18	25	35	45	23	32	73	79
RBC	13	18	30	36	17	22	71	78
RDW	14	15	32	33	18	19	71	72
TBIL	24	22	46	44	30	29	79	77
TC	21	26	43	50	27	31	80	87
TGL	—	—	—	—	—	—	—	—
TP	23	27	47	54	30	37	78	81
WBC	18	22	37	41	23	27	73	78
Overall	**21**	**25**	**40**	**48**	**25**	**32**	**76**	**80**

Table 14: Lead time for NORMA$_{\mathrm{RI}}$ abnormality detection relative to Pop$_{\mathrm{RI}}$. Per analyte and cohort: number of tests flagged by NORMA$_{\mathrm{RI}}$ alone (NORMA$_{\mathrm{RI}}$-only) and the subset later confirmed by Pop$_{\mathrm{RI}}$ (later population), with median lead time and interquartile range in hours.

	eICU-CRD				CHS			
Analyte	NORMA-only	Later Pop$_{RI}$	Median (h)	IQR (h)	NORMA-only	Later Pop$_{RI}$	Median (mo)	IQR (mo)
A1C	—	—	—	—	14,227	5,064	12.4	5.7–27.5
ALB	2,185	865	30.0	23.1–73.9	808,743	446,614	7.8	1.2–28.8
ALP	4,362	757	49.6	24.2–120.2	1,196,400	543,146	15.5	3.1–50.6
ALT	6,749	1,235	47.6	23.9–99.4	1,199,702	703,509	11.9	2.4–41.0
AST	7,720	1,317	47.4	23.9–95.5	1,195,620	551,419	10.0	1.6–38.0
BUN	25,578	7,039	28.0	23.1–71.3	568,660	402,995	7.6	1.1–29.2
CA	45,543	13,222	25.1	19.6–69.2	574,875	356,694	6.4	0.8–29.0
CL	24,313	8,477	34.7	23.0–72.8	44,929	36,351	4.9	0.5–22.7
CO2	52,641	26,365	25.6	19.3–64.5	96	80	2.9	0.3–8.7
CRE	40,464	8,744	26.0	21.6–67.7	1,070,918	687,844	7.5	1.4–27.1
DBIL	425	72	25.2	20.8–49.3	—	—	—	—
GLU	20,333	17,921	5.6	3.2–12.1	358,479	248,307	5.5	0.9–20.6
HCT	2,406	998	24.2	17.7–48.1	1,083,585	762,204	8.5	1.8–30.2
HDL	2	0	—	—	515,071	120,646	21.9	7.3–53.3
HGB	4,408	1,526	23.7	14.3–47.1	1,065,707	585,178	16.1	3.4–44.3
K	62,211	16,970	40.9	23.0–78.6	1,337,595	881,518	4.5	0.4–25.8
LDL	3	2	35.1	29.2–41.0	764,300	172,816	17.6	6.4–44.5
MCH	40,943	5,361	45.5	23.4–93.1	2,300,580	1,420,584	11.2	2.8–37.1
MCHC	38,761	11,733	25.1	22.5–49.9	840,255	776,047	7.6	1.6–29.9
MCV	19,132	1,789	48.2	23.7–120.7	4,464,421	1,941,232	18.8	4.1–62.8
MPV	101,702	4,805	37.5	23.2–72.5	2,030,046	8,370,452	20.0	4.8–53.3
NA	115,529	31,882	26.6	21.1–69.9	805,446	506,896	8.0	0.9–34.7
PLT	46,110	5,482	49.7	24.0–120.8	1,356,487	660,368	8.5	1.1–40.3
RBC	6,592	2,129	23.9	16.5–47.6	965	963	11.0	2.8–37.2
RDW	18,706	3,308	26.3	22.8–70.0	1,580,050	933,310	16.2	3.1–52.3
TBIL	11,136	1,721	47.4	23.8–95.5	2,966,137	1,268,891	26.4	6.0–68.5
TC	28	7	70.3	42.0–120.9	675,432	330,110	14.8	5.0–39.6
TGL	317	38	60.0	25.4–145.5	—	—	—	—
TP	11,311	2,592	34.8	22.9–74.1	962,840	530,403	6.9	0.9–28.6
WBC	31,253	8,381	35.3	23.3–72.7	2,183,898	1,402,701	7.4	1.0–33.0
Overall	740,863	184,738	34.8	23.1–72.6	50,235,864	24,646,342	9.3	1.7–35.9

Table 15: Proportional hazards analysis for in-hospital mortality in the eICU Collaborative Research Database. Per-analyte hazard ratios (with 95% confidence intervals) and concordance indices for Pop_{RI}, Per_{RI}, and NORMA_{RI} abnormality flags, adjusted for age and sex. Training and test event counts are shown.

			Pop_{RI}		Per_{RI}		NORMA_{RI}	
Analyte	N train (events)	N test (events)	HR [95% CI]	C	HR [95% CI]	C	HR [95% CI]	C
A1C	—	—	—	—	—	—	—	—
ALB	11151 (1560)	7435 (1040)	1.04 [0.75, 1.43]	0.544	1.01 [0.65, 1.57]	0.544	1.14 [0.78, 1.65]	0.544
ALP	7966 (1118)	5311 (746)	0.97 [0.86, 1.09]	0.534	1.09 [0.93, 1.28]	0.534	1.12 [0.99, 1.26]	0.535
ALT	7288 (976)	4859 (650)	0.98 [0.86, 1.11]	0.538	1.05 [0.89, 1.24]	0.539	0.99 [0.87, 1.13]	0.537
AST	7446 (912)	4964 (608)	1.68 [1.47, 1.93]	0.571	1.82 [1.52, 2.19]	0.562	1.61 [1.40, 1.86]	0.575
BUN	33379 (3378)	22253 (2252)	1.74 [1.58, 1.92]	0.573	1.84 [1.59, 2.14]	0.559	1.77 [1.59, 1.97]	0.571
CA	33080 (3474)	22054 (2316)	1.81 [1.64, 1.98]	0.577	1.68 [1.48, 1.91]	0.564	1.75 [1.58, 1.94]	0.571
CL	34320 (3560)	22881 (2374)	1.52 [1.40, 1.65]	0.572	1.69 [1.51, 1.89]	0.568	1.58 [1.44, 1.72]	0.574
CO2	40679 (4792)	27120 (3194)	1.94 [1.76, 2.13]	0.586	1.82 [1.60, 2.08]	0.566	1.90 [1.71, 2.11]	0.576
CRE	31723 (3152)	21149 (2102)	1.89 [1.74, 2.05]	0.602	1.97 [1.78, 2.17]	0.599	1.98 [1.81, 2.16]	0.604
DBIL	684 (122)	456 (82)	1.23 [0.81, 1.87]	0.546	1.07 [0.69, 1.65]	0.512	1.14 [0.75, 1.75]	0.537
GLU	52615 (5182)	35077 (3455)	2.14 [1.73, 2.65]	0.565	1.65 [1.26, 2.15]	0.563	2.19 [1.75, 2.74]	0.565
HCT	34809 (3553)	23206 (2368)	1.11 [0.89, 1.38]	0.558	1.45 [0.98, 2.15]	0.556	1.31 [1.01, 1.71]	0.558
HDL	—	—	—	—	—	—	—	—
HGB	35018 (3622)	23346 (2415)	1.01 [0.82, 1.23]	0.561	1.11 [0.83, 1.50]	0.561	1.07 [0.85, 1.35]	0.561
K	38603 (3943)	25736 (2628)	1.34 [1.26, 1.43]	0.563	1.31 [1.22, 1.41]	0.563	1.36 [1.27, 1.45]	0.564
LDL	—	—	—	—	—	—	—	—
MCH	28067 (2889)	18712 (1926)	0.99 [0.92, 1.07]	0.558	0.95 [0.88, 1.03]	0.558	0.99 [0.92, 1.06]	0.558
MCHC	29987 (3063)	19992 (2042)	0.85 [0.78, 0.92]	0.563	0.87 [0.78, 0.97]	0.558	0.89 [0.82, 0.98]	0.561
MCV	29488 (3017)	19659 (2011)	1.26 [1.16, 1.37]	0.559	1.35 [1.25, 1.47]	0.564	1.35 [1.25, 1.45]	0.569
MPV	21532 (2191)	14355 (1461)	0.95 [0.84, 1.07]	0.568	1.17 [1.08, 1.28]	0.575	1.20 [1.02, 1.40]	0.571
NA	36226 (3839)	24152 (2559)	1.22 [1.14, 1.31]	0.551	1.37 [1.25, 1.50]	0.557	1.22 [1.11, 1.35]	0.552
PLT	30955 (3214)	20638 (2142)	2.50 [2.32, 2.68]	0.654	1.77 [1.58, 1.99]	0.570	1.67 [1.54, 1.82]	0.579
RBC	31071 (3221)	20714 (2148)	1.05 [0.88, 1.25]	0.562	1.07 [0.85, 1.34]	0.561	1.10 [0.90, 1.33]	0.561
RDW	28255 (2852)	18838 (1901)	1.85 [1.65, 2.07]	0.601	2.23 [1.89, 2.63]	0.589	1.91 [1.68, 2.16]	0.595
TBIL	7465 (930)	4978 (620)	1.39 [1.22, 1.59]	0.552	1.28 [1.12, 1.47]	0.550	1.38 [1.22, 1.58]	0.552
TC	—	—	—	—	—	—	—	—
TGL	186 (37)	124 (24)	1.09 [0.54, 2.18]	0.594	1.03 [0.43, 2.48]	0.590	1.22 [0.59, 2.49]	0.598
TP	8881 (1307)	5921 (871)	1.67 [1.44, 1.93]	0.576	1.68 [1.37, 2.06]	0.554	1.66 [1.42, 1.95]	0.562
WBC	30250 (2917)	20167 (1945)	2.39 [2.17, 2.63]	0.619	2.48 [2.16, 2.85]	0.598	2.41 [2.17, 2.67]	0.617

Table 16: Proportional hazards analysis for acute kidney injury in the eICU Collaborative Research Database. Per-analyte hazard ratios (with 95% confidence intervals) and concordance indices for Pop_{RI}, Per_{RI}, and NORMA_{RI} abnormality flags, adjusted for age and sex.

			Pop_{RI}		Per_{RI}		NORMA_{RI}	
Analyte	N train (events)	N test (events)	HR [95% CI]	C	HR [95% CI]	C	HR [95% CI]	C
A1C	—	—	—	—	—	—	—	—
ALB	11121 (2490)	7414 (1660)	1.28 [1.01, 1.63]	0.516	1.45 [1.03, 2.03]	0.515	1.36 [1.04, 1.79]	0.515
ALP	7949 (1702)	5300 (1134)	1.25 [1.14, 1.37]	0.544	1.12 [0.99, 1.26]	0.516	1.24 [1.12, 1.36]	0.534
ALT	7267 (1472)	4846 (982)	1.08 [0.97, 1.19]	0.504	0.94 [0.82, 1.07]	0.505	1.06 [0.95, 1.18]	0.504
AST	7429 (1464)	4954 (976)	1.19 [1.08, 1.32]	0.529	1.07 [0.95, 1.21]	0.504	1.14 [1.03, 1.27]	0.522
BUN	33279 (4964)	22187 (3310)	1.89 [1.76, 2.04]	0.566	1.63 [1.47, 1.82]	0.535	1.73 [1.60, 1.87]	0.555
CA	32980 (5407)	21987 (3604)	1.43 [1.34, 1.53]	0.541	1.40 [1.28, 1.52]	0.533	1.39 [1.29, 1.49]	0.539
CL	34213 (5471)	22809 (3647)	1.43 [1.34, 1.52]	0.540	1.24 [1.14, 1.34]	0.525	1.43 [1.34, 1.52]	0.538
CO2	40566 (6164)	27044 (4109)	1.40 [1.31, 1.50]	0.534	1.35 [1.23, 1.49]	0.524	1.33 [1.24, 1.44]	0.528
CRE	31638 (4454)	21092 (2969)	1.77 [1.66, 1.89]	0.576	1.55 [1.44, 1.66]	0.555	1.59 [1.49, 1.69]	0.560
DBIL	682 (211)	456 (141)	1.32 [0.97, 1.79]	0.520	1.32 [0.94, 1.86]	0.520	1.40 [1.01, 1.93]	0.523
GLU	52476 (6866)	34984 (4577)	1.06 [0.94, 1.19]	0.519	0.96 [0.82, 1.13]	0.518	1.04 [0.92, 1.17]	0.519
HCT	34716 (5015)	23144 (3343)	1.91 [1.55, 2.36]	0.524	2.59 [1.78, 3.79]	0.521	1.97 [1.56, 2.49]	0.524
HDL	—	—	—	—	—	—	—	—
HGB	34923 (5009)	23282 (3339)	1.97 [1.62, 2.40]	0.527	2.18 [1.61, 2.96]	0.524	2.01 [1.61, 2.50]	0.526
K	38496 (5647)	25664 (3764)	1.40 [1.33, 1.48]	0.557	1.33 [1.26, 1.41]	0.544	1.39 [1.32, 1.47]	0.554
LDL	—	—	—	—	—	—	—	—
MCH	27981 (4313)	18654 (2875)	1.12 [1.05, 1.19]	0.528	1.08 [1.01, 1.16]	0.527	1.14 [1.07, 1.21]	0.528
MCHC	29898 (4621)	19933 (3080)	1.17 [1.10, 1.25]	0.531	1.14 [1.05, 1.25]	0.529	1.13 [1.05, 1.21]	0.530
MCV	29400 (4528)	19601 (3018)	1.12 [1.04, 1.21]	0.523	1.02 [0.96, 1.08]	0.522	1.06 [1.00, 1.13]	0.523
MPV	21459 (3144)	14307 (2096)	1.30 [1.18, 1.43]	0.535	1.07 [1.00, 1.15]	0.523	1.10 [0.98, 1.24]	0.526
NA	36123 (5599)	24082 (3733)	1.13 [1.07, 1.19]	0.527	1.06 [0.99, 1.13]	0.528	1.01 [0.94, 1.08]	0.525
PLT	30866 (4735)	20578 (3156)	1.38 [1.30, 1.46]	0.543	1.03 [0.95, 1.12]	0.517	1.15 [1.08, 1.22]	0.524
RBC	30981 (4736)	20654 (3158)	1.75 [1.50, 2.04]	0.520	1.71 [1.40, 2.09]	0.521	1.78 [1.50, 2.11]	0.521
RDW	28174 (4340)	18783 (2894)	1.73 [1.60, 1.87]	0.571	1.65 [1.50, 1.83]	0.553	1.71 [1.58, 1.86]	0.568
TBIL	7447 (1482)	4966 (988)	1.15 [1.04, 1.28]	0.526	1.11 [1.00, 1.23]	0.527	1.12 [1.01, 1.24]	0.532
TC	20 (4)	14 (2)	0.52 [0.07, 3.77]	0.640	—	0.600	0.89 [0.08, 10.23]	0.640
TGL	186 (54)	124 (36)	1.18 [0.67, 2.09]	0.551	0.97 [0.47, 2.01]	0.515	1.02 [0.58, 1.82]	0.525
TP	8860 (1890)	5908 (1260)	1.28 [1.15, 1.43]	0.517	1.34 [1.15, 1.55]	0.510	1.31 [1.16, 1.48]	0.518
WBC	30162 (4532)	20108 (3021)	1.27 [1.19, 1.35]	0.537	1.17 [1.08, 1.26]	0.524	1.28 [1.20, 1.36]	0.531

Table 17: Proportional hazards analysis for sepsis in the eICU Collaborative Research Database. Per-analyte hazard ratios (with 95% confidence intervals) and concordance indices for Pop_{RI}, Per_{RI}, and NORMA_{RI} abnormality flags, adjusted for age and sex.

			Pop_{RI}		Per_{RI}		NORMA_{RI}	
Analyte	N train (events)	N test (events)	HR [95% CI]	C	HR [95% CI]	C	HR [95% CI]	C
A1C	—	—	—	—	—	—	—	—
ALB	11106 (2754)	7405 (1836)	2.27 [1.72, 3.00]	0.519	2.42 [1.64, 3.56]	0.518	2.24 [1.66, 3.02]	0.518
ALP	7938 (2020)	5292 (1347)	1.17 [1.07, 1.28]	0.544	0.98 [0.88, 1.10]	0.503	1.12 [1.02, 1.22]	0.532
ALT	7256 (1877)	4838 (1252)	1.01 [0.92, 1.11]	0.510	1.01 [0.90, 1.13]	0.510	0.99 [0.90, 1.09]	0.512
AST	7415 (1904)	4944 (1270)	1.08 [0.98, 1.18]	0.502	0.96 [0.87, 1.07]	0.517	1.08 [0.98, 1.18]	0.495
BUN	33240 (5690)	22161 (3793)	1.22 [1.15, 1.30]	0.533	1.08 [0.99, 1.17]	0.527	1.16 [1.09, 1.24]	0.527
CA	32941 (5802)	21962 (3868)	1.53 [1.44, 1.63]	0.549	1.56 [1.43, 1.71]	0.538	1.51 [1.40, 1.62]	0.544
CL	34176 (5909)	22785 (3939)	1.34 [1.26, 1.41]	0.539	1.29 [1.20, 1.39]	0.528	1.32 [1.24, 1.40]	0.536
CO2	40521 (6798)	27015 (4532)	1.26 [1.18, 1.34]	0.540	1.17 [1.07, 1.27]	0.532	1.19 [1.12, 1.28]	0.534
CRE	31595 (5314)	21064 (3542)	1.28 [1.21, 1.35]	0.547	1.21 [1.14, 1.29]	0.536	1.21 [1.15, 1.28]	0.537
DBIL	682 (186)	456 (125)	0.98 [0.72, 1.33]	0.513	1.01 [0.72, 1.41]	0.515	1.09 [0.79, 1.50]	0.522
GLU	52429 (7603)	34953 (5068)	0.98 [0.88, 1.09]	0.527	0.93 [0.80, 1.08]	0.527	0.96 [0.86, 1.08]	0.527
HCT	34671 (5744)	23114 (3829)	1.50 [1.26, 1.79]	0.528	1.75 [1.31, 2.34]	0.527	1.47 [1.22, 1.78]	0.527
HDL	—	—	—	—	—	—	—	—
HGB	34876 (5787)	23252 (3858)	1.50 [1.28, 1.76]	0.531	1.53 [1.22, 1.93]	0.527	1.50 [1.26, 1.79]	0.529
K	38461 (6288)	25642 (4192)	1.48 [1.40, 1.55]	0.560	1.32 [1.25, 1.40]	0.541	1.41 [1.34, 1.48]	0.551
LDL	—	—	—	—	—	—	—	—
MCH	27943 (5010)	18630 (3340)	1.20 [1.14, 1.27]	0.533	1.11 [1.04, 1.18]	0.525	1.19 [1.13, 1.26]	0.533
MCHC	29860 (5363)	19908 (3575)	1.32 [1.24, 1.41]	0.547	1.30 [1.20, 1.41]	0.529	1.31 [1.23, 1.40]	0.541
MCV	29362 (5220)	19575 (3480)	1.24 [1.16, 1.33]	0.529	1.12 [1.05, 1.18]	0.510	1.22 [1.15, 1.29]	0.520
MPV	21432 (3423)	14289 (2282)	1.27 [1.16, 1.39]	0.520	1.05 [0.98, 1.12]	0.513	1.04 [0.93, 1.16]	0.514
NA	36085 (6030)	24058 (4020)	1.05 [1.00, 1.11]	0.519	1.00 [0.93, 1.06]	0.519	0.98 [0.92, 1.05]	0.519
PLT	30826 (5403)	20552 (3603)	1.16 [1.10, 1.23]	0.532	1.03 [0.95, 1.11]	0.526	1.07 [1.01, 1.13]	0.530
RBC	30939 (5515)	20626 (3677)	1.38 [1.22, 1.57]	0.522	1.47 [1.24, 1.74]	0.518	1.36 [1.19, 1.56]	0.519
RDW	28135 (5022)	18758 (3349)	1.83 [1.70, 1.96]	0.557	1.67 [1.52, 1.83]	0.535	1.77 [1.64, 1.91]	0.545
TBIL	7437 (1878)	4959 (1252)	1.02 [0.93, 1.12]	0.517	0.98 [0.89, 1.08]	0.511	1.01 [0.92, 1.11]	0.515
TC	20 (6)	14 (5)	0.71 [0.13, 3.89]	0.425	—	0.375	0.76 [0.08, 7.71]	0.425
TGL	186 (63)	124 (42)	0.95 [0.56, 1.59]	0.409	1.28 [0.63, 2.63]	0.439	0.89 [0.53, 1.50]	0.405
TP	8848 (2278)	5899 (1518)	1.07 [0.97, 1.18]	0.529	1.08 [0.95, 1.22]	0.527	1.11 [1.00, 1.24]	0.531
WBC	30127 (5170)	20086 (3447)	1.48 [1.39, 1.57]	0.545	1.28 [1.19, 1.38]	0.523	1.39 [1.31, 1.48]	0.536

Table 18: Proportional hazards analysis for prolonged ICU stay in the eICU Collaborative Research Database. Prolonged ICU stay is defined as ICU stay greater than 7 days. Per-analyte hazard ratios (with 95% confidence intervals) and concordance indices for Pop_{RI}, Per_{RI}, and NORMA_{RI} abnormality flags, adjusted for age and sex.

			Pop_{RI}		Per_{RI}		NORMA_{RI}	
Analyte	N train (events)	N test (events)	HR [95% CI]	C	HR [95% CI]	C	HR [95% CI]	C
A1C	—	—	—	—	—	—	—	—
ALB	11140 (3587)	7428 (2392)	0.75 [0.61, 0.93]	0.535	0.70 [0.51, 0.96]	0.536	0.77 [0.60, 0.97]	0.535
ALP	7956 (2821)	5305 (1881)	0.68 [0.63, 0.73]	0.551	0.84 [0.76, 0.92]	0.533	0.73 [0.68, 0.79]	0.548
ALT	7276 (2596)	4851 (1731)	0.82 [0.76, 0.88]	0.533	0.87 [0.78, 0.96]	0.520	0.85 [0.78, 0.92]	0.527
AST	7437 (2658)	4958 (1772)	0.86 [0.80, 0.93]	0.545	0.98 [0.89, 1.07]	0.534	0.87 [0.81, 0.94]	0.545
BUN	33336 (7040)	22225 (4694)	0.66 [0.62, 0.71]	0.550	0.76 [0.70, 0.83]	0.535	0.67 [0.63, 0.72]	0.548
CA	33039 (6996)	22026 (4664)	0.79 [0.75, 0.84]	0.541	0.77 [0.71, 0.83]	0.537	0.75 [0.71, 0.80]	0.544
CL	34278 (7202)	22853 (4801)	0.74 [0.70, 0.78]	0.552	0.84 [0.79, 0.90]	0.538	0.75 [0.71, 0.80]	0.548
CO2	40636 (7193)	27092 (4795)	0.63 [0.58, 0.68]	0.547	0.72 [0.66, 0.80]	0.540	0.61 [0.56, 0.66]	0.545
CRE	31684 (6787)	21123 (4525)	0.68 [0.65, 0.72]	0.566	0.66 [0.62, 0.70]	0.562	0.69 [0.66, 0.73]	0.566
DBIL	684 (328)	456 (218)	1.02 [0.81, 1.29]	0.512	0.97 [0.75, 1.25]	0.482	0.95 [0.75, 1.22]	0.461
GLU	52561 (7373)	35041 (4915)	0.44 [0.36, 0.53]	0.537	0.53 [0.41, 0.70]	0.535	0.47 [0.38, 0.57]	0.536
HCT	34765 (7128)	23178 (4753)	0.53 [0.46, 0.61]	0.541	0.64 [0.51, 0.81]	0.537	0.53 [0.45, 0.62]	0.539
HDL	—	—	—	—	—	—	—	—
HGB	34972 (7128)	23316 (4752)	0.50 [0.43, 0.58]	0.543	0.51 [0.42, 0.62]	0.537	0.52 [0.44, 0.61]	0.542
K	38559 (7280)	25707 (4854)	0.77 [0.73, 0.80]	0.553	0.81 [0.77, 0.85]	0.543	0.76 [0.73, 0.80]	0.554
LDL	—	—	—	—	—	—	—	—
MCH	28026 (6451)	18684 (4301)	0.92 [0.87, 0.97]	0.537	0.86 [0.81, 0.91]	0.544	0.83 [0.79, 0.87]	0.541
MCHC	29944 (6867)	19964 (4578)	0.67 [0.63, 0.71]	0.559	0.67 [0.62, 0.72]	0.552	0.65 [0.61, 0.69]	0.559
MCV	29445 (6783)	19631 (4523)	0.88 [0.83, 0.94]	0.539	0.97 [0.92, 1.02]	0.540	0.83 [0.80, 0.88]	0.546
MPV	21492 (4995)	14328 (3330)	0.85 [0.78, 0.92]	0.535	0.98 [0.92, 1.04]	0.529	0.93 [0.84, 1.03]	0.532
NA	36186 (7220)	24125 (4814)	0.74 [0.71, 0.78]	0.560	0.80 [0.75, 0.85]	0.543	0.76 [0.71, 0.82]	0.549
PLT	30913 (7030)	20609 (4686)	0.90 [0.86, 0.94]	0.534	1.22 [1.14, 1.30]	0.542	0.96 [0.91, 1.01]	0.537
RBC	31027 (7105)	20686 (4737)	0.55 [0.49, 0.62]	0.549	0.53 [0.45, 0.63]	0.541	0.54 [0.47, 0.62]	0.547
RDW	28216 (6496)	18812 (4331)	0.67 [0.63, 0.71]	0.557	0.70 [0.65, 0.76]	0.546	0.67 [0.63, 0.72]	0.554
TBIL	7456 (2547)	4972 (1699)	0.86 [0.79, 0.93]	0.537	0.90 [0.83, 0.98]	0.542	0.87 [0.81, 0.95]	0.534
TC	20 (8)	14 (5)	0.67 [0.11, 4.11]	0.600	—	—	0.20 [0.00, 12.22]	0.500
TGL	186 (134)	124 (90)	0.82 [0.57, 1.18]	0.527	0.74 [0.46, 1.19]	0.533	0.91 [0.63, 1.31]	0.528
TP	8871 (3098)	5914 (2065)	1.03 [0.95, 1.12]	0.524	0.97 [0.87, 1.09]	0.527	0.98 [0.90, 1.07]	0.527
WBC	30209 (7003)	20140 (4668)	0.75 [0.71, 0.79]	0.557	0.79 [0.74, 0.85]	0.545	0.74 [0.69, 0.78]	0.552

Table 19: Proportional hazards analysis for all-cause mortality in Clalit Health Services cohort. Per-analyte hazard ratios (with 95% confidence intervals) and concordance indices for Pop_{RI}, Per_{RI}, and NORMA_{RI} abnormality flags, adjusted for age and sex. Training and test event counts are shown.

			Pop_{RI}		Per_{RI}		NORMA_{RI}	
Analyte	N train (events)	N test (events)	HR [95% CI]	C	HR [95% CI]	C	HR [95% CI]	C
A1C	9517 (3118)	6345 (2079)	0.84 [0.65, 1.09]	0.689	0.58 [0.54, 0.63]	0.699	0.82 [0.65, 1.03]	0.688
ALB	13612 (4221)	9076 (2815)	4.15 [3.83, 4.49]	0.785	1.21 [1.10, 1.33]	0.740	3.11 [2.80, 3.46]	0.757
ALP	20262 (5615)	13508 (3743)	2.05 [1.94, 2.17]	0.760	0.96 [0.87, 1.05]	0.741	2.01 [1.86, 2.18]	0.749
ALT	25254 (6667)	16836 (4445)	2.08 [1.95, 2.22]	0.761	0.94 [0.86, 1.03]	0.749	1.35 [1.27, 1.43]	0.752
AST	25296 (6662)	16864 (4442)	1.85 [1.76, 1.94]	0.758	1.02 [0.94, 1.12]	0.744	1.55 [1.47, 1.65]	0.750
BUN	26980 (7187)	17987 (4792)	2.82 [2.53, 3.15]	0.747	0.81 [0.75, 0.88]	0.739	2.58 [2.31, 2.87]	0.746
CA	15946 (4688)	10631 (3125)	1.93 [1.81, 2.07]	0.754	1.08 [0.99, 1.17]	0.740	1.88 [1.73, 2.05]	0.749
CL	846 (307)	564 (205)	3.57 [1.67, 7.61]	0.767	1.90 [0.97, 3.71]	0.760	4.10 [1.82, 9.27]	0.755
CO2	—	—	—	—	—	—	—	—
CRE	29331 (7624)	19555 (5083)	3.00 [2.82, 3.18]	0.770	1.04 [0.97, 1.12]	0.744	3.30 [3.07, 3.54]	0.768
DBIL	—	—	—	—	—	—	—	—
GLU	28928 (7343)	19286 (4895)	1.60 [1.39, 1.85]	0.749	0.77 [0.70, 0.84]	0.748	1.61 [1.38, 1.86]	0.749
HCT	31026 (7576)	20685 (5051)	3.60 [3.15, 4.12]	0.774	0.79 [0.73, 0.86]	0.765	3.36 [2.86, 3.94]	0.771
HDL	20800 (5296)	13867 (3530)	1.68 [1.56, 1.81]	0.733	0.73 [0.67, 0.81]	0.726	1.72 [1.57, 1.89]	0.730
HGB	31020 (7578)	20680 (5052)	3.70 [3.32, 4.12]	0.779	0.83 [0.77, 0.90]	0.764	3.72 [3.24, 4.26]	0.772
K	25819 (7121)	17213 (4748)	1.63 [1.53, 1.73]	0.748	1.07 [0.99, 1.15]	0.742	1.69 [1.56, 1.83]	0.746
LDL	23706 (6093)	15804 (4062)	0.81 [0.77, 0.85]	0.735	0.74 [0.69, 0.79]	0.736	0.87 [0.82, 0.92]	0.734
MCH	28147 (6984)	18765 (4656)	2.04 [1.89, 2.20]	0.767	0.98 [0.91, 1.07]	0.760	2.18 [1.90, 2.50]	0.763
MCHC	31009 (7580)	20673 (5053)	1.20 [0.89, 1.61]	0.767	0.96 [0.90, 1.03]	0.767	0.99 [0.76, 1.28]	0.767
MCV	30848 (7546)	20566 (5031)	1.70 [1.62, 1.78]	0.774	0.90 [0.83, 0.97]	0.762	1.95 [1.77, 2.16]	0.766
MPV	28679 (7094)	19120 (4729)	1.28 [1.22, 1.35]	0.765	0.93 [0.87, 0.99]	0.762	0.10 [0.04, 0.23]	0.762
NA	25977 (7133)	17318 (4756)	2.17 [2.04, 2.31]	0.753	1.03 [0.95, 1.12]	0.738	2.02 [1.88, 2.16]	0.750
PLT	30976 (7564)	20652 (5043)	1.98 [1.89, 2.08]	0.779	0.92 [0.83, 1.01]	0.765	1.88 [1.77, 1.98]	0.775
RBC	30946 (7556)	20631 (5038)	—	—	0.78 [0.71, 0.85]	0.764	—	—
RDW	30874 (7528)	20583 (5018)	3.09 [2.82, 3.39]	0.774	0.90 [0.83, 0.97]	0.762	2.23 [1.96, 2.52]	0.765
TBIL	13506 (3888)	9005 (2593)	1.61 [1.50, 1.72]	0.744	1.13 [1.04, 1.22]	0.737	1.09 [0.98, 1.21]	0.738
TC	25805 (6534)	17204 (4357)	0.93 [0.87, 0.99]	0.739	0.80 [0.75, 0.86]	0.739	0.96 [0.90, 1.03]	0.739
TGL	—	—	—	—	—	—	—	—
TP	13638 (4217)	9092 (2812)	2.23 [2.08, 2.38]	0.756	1.07 [0.98, 1.16]	0.736	2.73 [2.46, 3.04]	0.745
WBC	31053 (7583)	20702 (5055)	1.86 [1.75, 1.97]	0.773	0.98 [0.89, 1.07]	0.763	1.90 [1.75, 2.07]	0.768